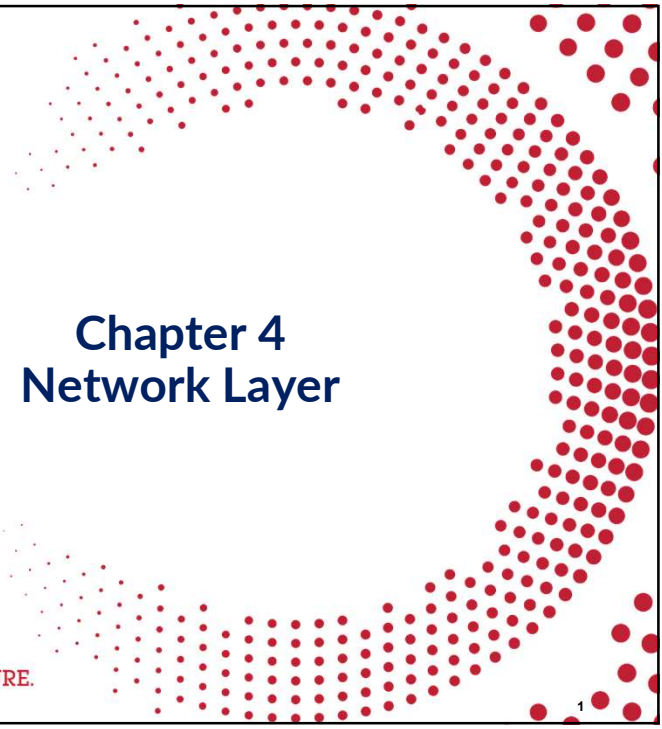


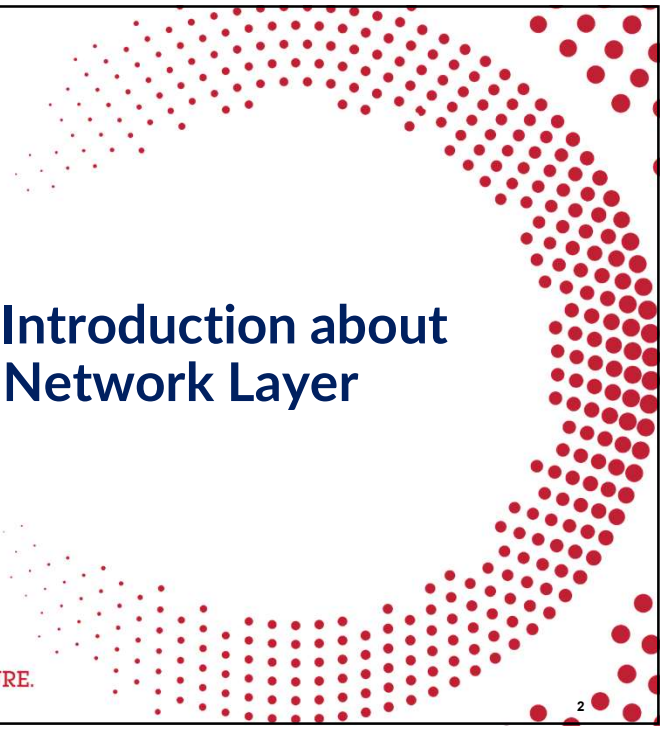


Chapter 4 Network Layer

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1

1



4.1 Introduction about Network Layer

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2

2

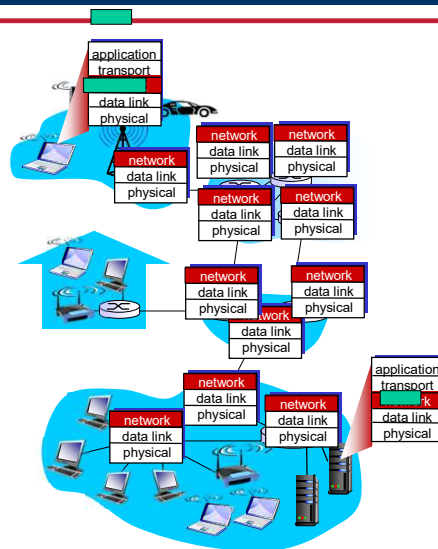
Network layer in the TCP/IP architecture

Application (HTTP, Mail, ...) Transport (UDP, TCP ...) Network (IP, ICMP...) Datalink (Ethernet, ADSL...) Physical (bits...)	Support network applications Control data transmission between processes at the application layer Controlling data transmission to another network Facilitating communication for subsequent components on the same network Transmitting and receiving bit streams on the physical transmission line
--	--

3

Network layer

- Data transmission from host to host
- Implemented on every end system and router
- Transmission unit: datagram
- Sender: receives data from the transport layer, encapsulates it
- Receiver: opens the package, transfers the data portion in the payload to the transport layer
- Router: routes and forwards packets



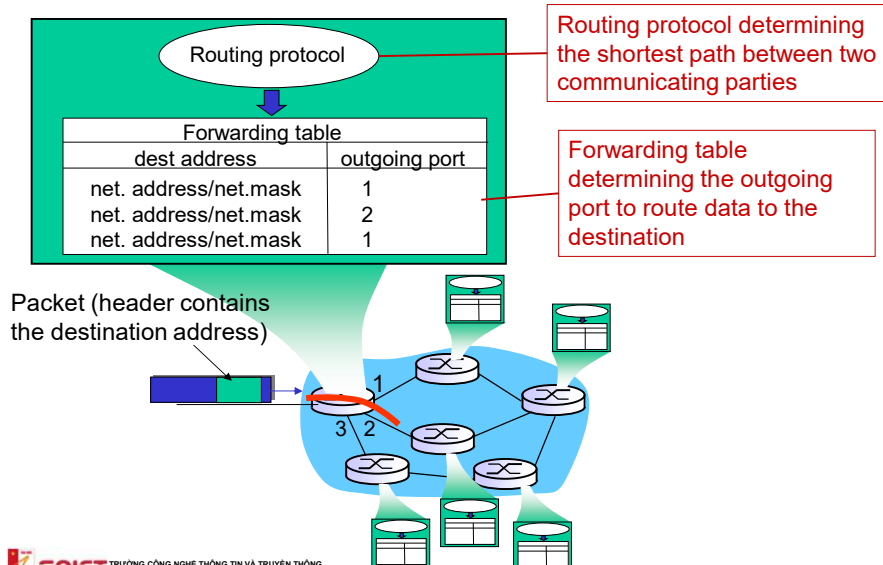
Hình ảnh từ: "Computer Networking: A Top Down Approach", Jim Kurose

4

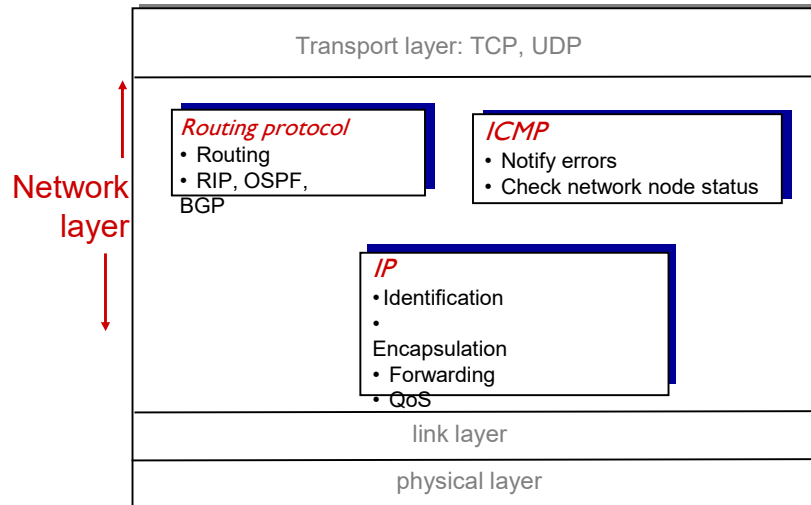
Main functionalities

- **Routing:** Finding a path to send data from source to destination
- **Forwarding:** Sending a packet to the outgoing port along the established route
- **Addressing:** Identifying addresses for network nodes
- **Encapsulating data:** Packaging data for transmission
- **Quality of Service (QoS):** Ensuring appropriate parameters of the transmission path for each service

Routing and forwarding



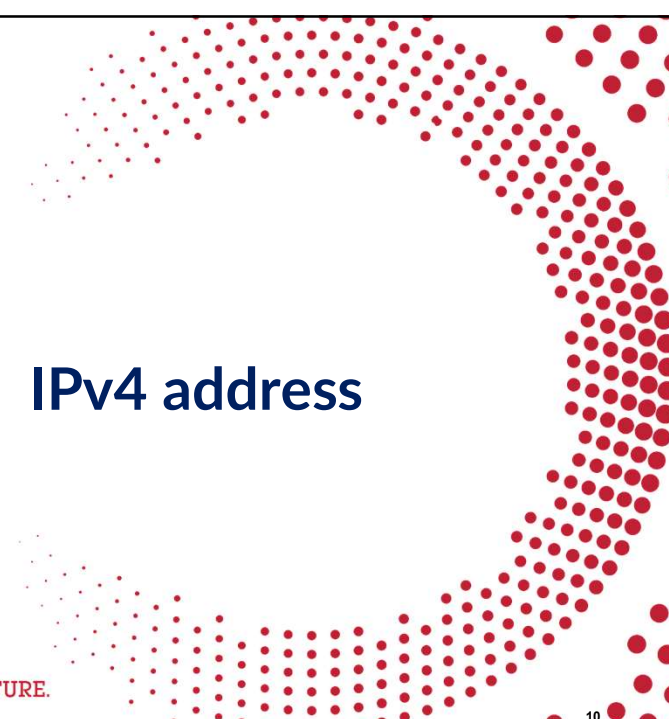
Network layer protocols



4.2. Internet Protocol v4

Basic characteristics of IPv4

- Connectionless-oriented protocol
- Unreliable / Fast
 - Data transmission using 'best effort' method
 - IP lacks error recovery mechanisms
 - When needed, applications will use upper-layer services to ensure reliability (TCP)
- It is a routed protocol
- Requires routing protocols to pre-determine paths for data.
- Currently, there are 2 versions: IPv4 and IPv6.

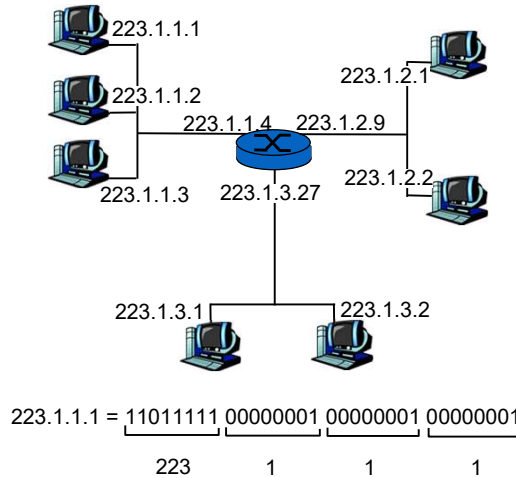


IPv4 address

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IP address (IPv4)

- **IP address:** A 32-bit number identifying *uniquely* a *network interface*
- **Interface:**
 - Router's typically have multiple interfaces
 - host may have multiple interfaces
 - IP addresses associated with interface, not host, router

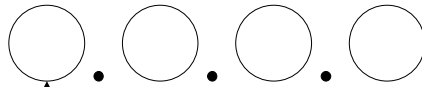


IPv4 address allocation

- **Static IP allocation:**
 - Windows: Control Panel → Network → Configuration → TCP/IP → Properties
 - Linux: /etc/network/interfaces
- **Dynamic allocation: DHCP- Dynamic Host Configuration Protocol**

IPv4 representation

- Decimal representation with dots



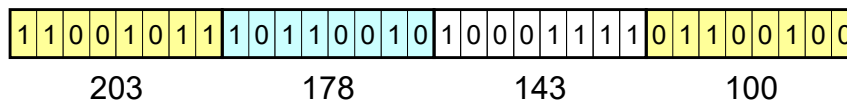
8 bits
0 – 255 integer

Example:

203.178.136.63	o
259.12.49.192	x
133.27.4.27	o

Use 4 x 8 bits describing a 32 bits address

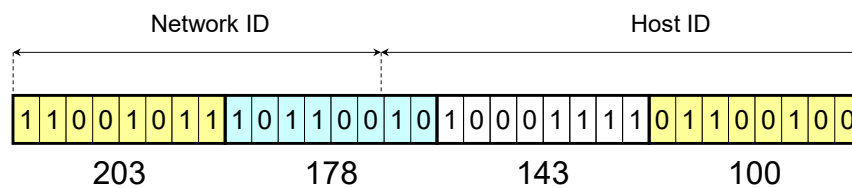
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Host address, network address

- IP address contains two parts
 - Host ID – identify a host in a network
 - Network ID – identify a network



- How to know which bits belong to network ID or host ID parts?
 - Use classful IP address
 - Use classless IP address– CIDR

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Address types

- Network Address:
 - Identification for a network
 - All bits in the HostID part are set to 0
 - 0.0.0.0: Network-wide address, representing any network
- Broadcast Address
 - Address used to send data to all stations in the network
 - All bits in the HostID part are set to 1
 - 255.255.255.255: Broadcast address within a network
- Unicast Address
 - Assign to a network interface
- Multicast address: Identification for a group

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Classful addressing for IPv4 addresses

	8bits		8bits		8bits		8bits	
Class A	0	7bit	H	H	H			
Class B	1	0	6bit	N	H	H		
Class C	1	1	0	5bit	N	N	H	
Class D	1	1	1	0	Multicast			
Class E	1	1	1	1	Reserve for future use			

	# of network	# of hosts
Class A	128	2^{24}
Class B	16384	65536
Class C	2^{21}	256

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Exercises

- Determine which classes do these IP addresses belong to:
 - 10.10.10.9
 - 192.168.70.5
 - 129.60.4.7

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Limitation of classful IP address

- Inefficient use of addressing space
 - Hard classification of addressing space into classes (A, B, C, D, E) makes it is difficult to use all the address space

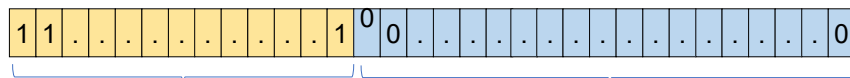
Solution...

- CIDR: **C**lassless **I**nter **D**omain **R**outing
 - Network ID part will have variable length.
 - Length of Network ID part is specified in Network mask
 - Address notation: **a.b.c.d/x**, where x (mask) the number of bit of Network ID part.

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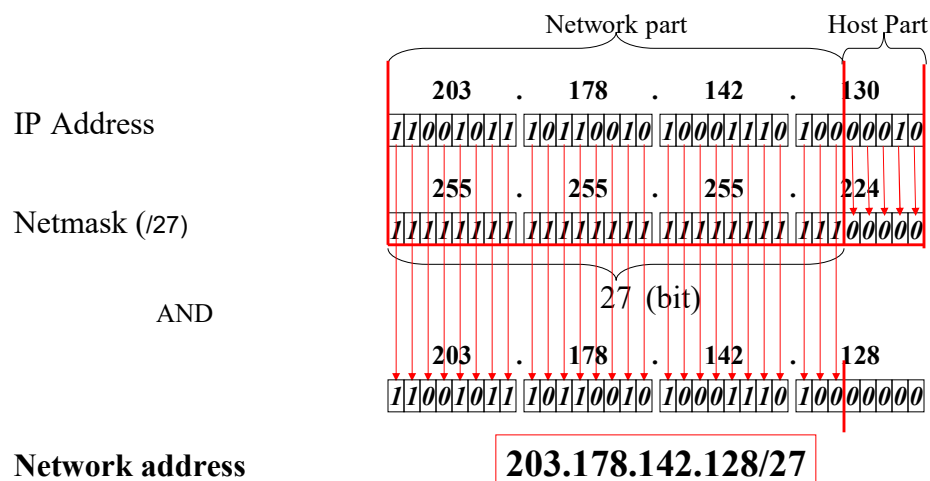
Network mask

- Network mask divides the IP address into two parts
 - Part corresponding to Host ID
 - Part corresponding to Network ID
- Representation:
 - Prefix: /n. Example: /27
 - n: Number of bits of NetworkID
 - Binary representation:



- Example: /27 = 11111111 11111111 11111111 11100000
- Decimal representation with dot: 255.255.255.224

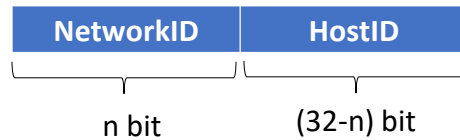
Calculation of network address



Network mask and network size

- Network mask: /n
- Network size: The maximum nodes in a network

- IP Address:



- The number of addresses for hosts (Unicast Address):

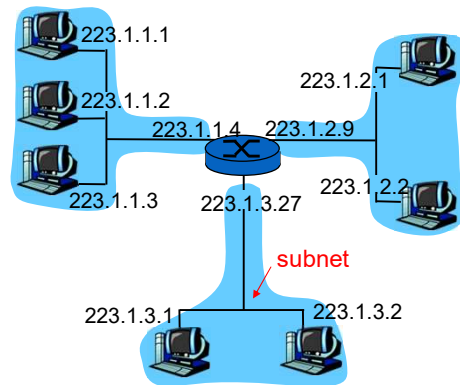
$$2^{32-n} - 2$$

Exercise: IP address and network mask

- Which of the following IP addresses are host address, network address, broadcast address?
 - (1) 203.178.142.128 /25
 - (2) 203.178.142.128 /24
 - (3) 203.178.142.127 /25
 - (4) 203.178.142.127 /24
- Attn: With CIDR addressing, IP address should always coming with a network mask

Subnet

- Subnet is a part of a network
 - Hosts of a subnet communicate directly without reaching to layer 3.
 - Usually is one department of an organization
- Design question: How to assign addresses of a network to subnets
 - Use a longer netmask



A network with 3 subnets.

Example: Divide into 2 subnets

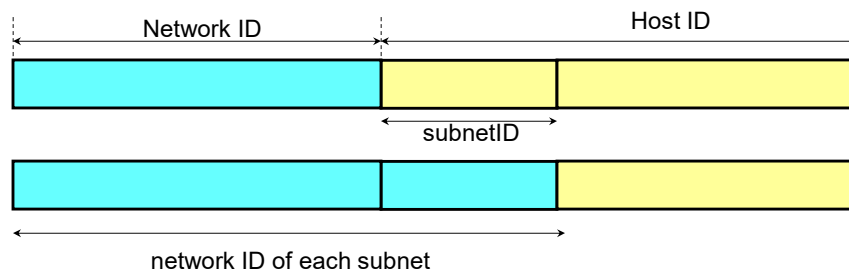
11001000 00010111 00010000 00000000
200. 23. 16. 0 /24



11001000 00010111 00010000 00000000
200. 23. 16. 0 /25
11001000 00010111 00010000 10000000
200. 23. 16. 128 /25

Principle

- Divide a IP range into sub-ranges of equal size
- Take some bits from HostID part to distinguish subnets
 - each subnet contains IP addresses with a fixed values of subnet ID.



Exercise: Dividing into subnets

- Given IP addresses in the range 200.23.16.0/24

1) Need to organize into 8 subnets

- Address of each subnetwork? Mask? Number of hosts/network
- 200.23.16.0 /27

2) General question: Need to create N subnets.
Network address? Mask?

- Each network contains 14 hosts /28
- Each network contains 30 hosts → /27
- Each network contains 31 hosts /26
- Each network contains 70 hosts /25

Addressing space of IPv4

- In theory
 - All between 0.0.0.0 ~ 255.255.255.255
 - Some special IP address ([RFC1918](#))

Private address	10.0.0.0/8
	172.16.0.0/12
	192.168.0.0/16
Loopback address	127.0.0.0
Multicast address	224.0.0.0
	~ 239.255.255.255

- Self assigned IP address: 169.254.0.0/16 (self-configuration)

Attention about IP

- Currently IPv4: 32 bits
 - 133.113.215.10 (IPv4)
- IPv6 is also widely used: 128bits
 - 2001:200:0:8803::53 (IPv6)
 - Fix 64 first bit for subnet ID, 64 last bit belongs to interface ID.
 - Security feature is integrated

Management of public IP addresses

- Internet Corporation for Assigned Names and Numbers (ICANN): Manages the entire IP address resources
- Regional Internet Registries: Manage IP addresses by regions (Asia-Pacific, Europe and the Middle East, Africa, North America, South America)
- National management agencies
 - Vietnam: VNNIC
- Internet Service Provider (ISP)
- Agencies, organizations
- Examples: ICANN → APNIC → VNNIC → HUST



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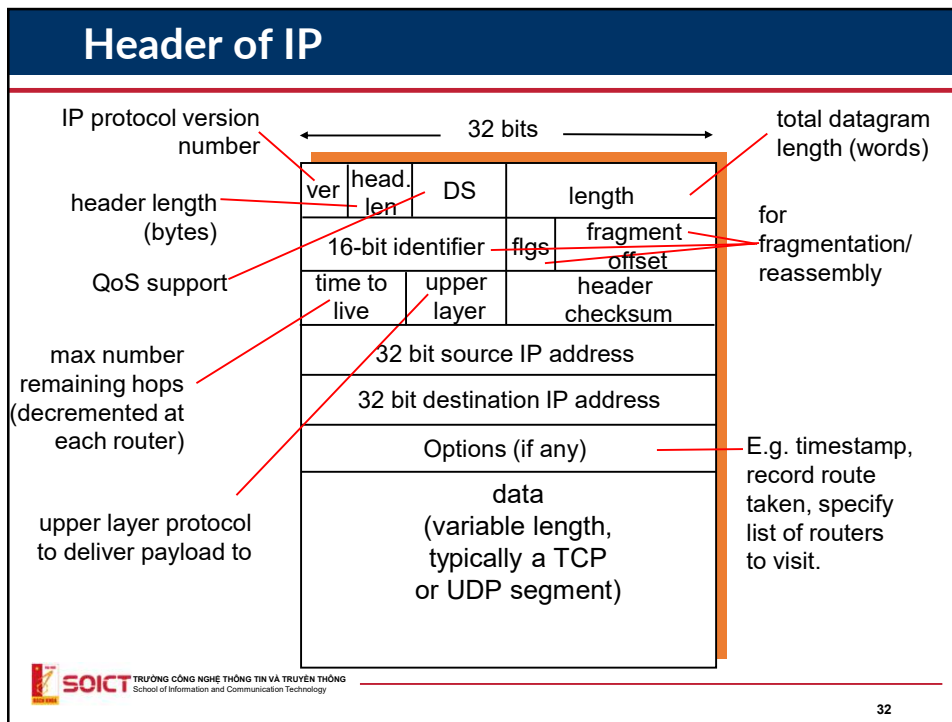
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4.3. IP packet's format

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IP header (1)

- Version (4 bits)
 - IPv4
 - IPv6
- Header length: 4bits
 - In word unit (4 bytes)
 - Min: 5
 - Max: 60
- DS (Differentiated Service : 8bits)
 - Old name: Type of Service
 - Used for QoS management by some router
 - Diffserv

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IP header (2)

- Length: total length including header (16 bits)
 - In bytes unit
 - Max: 65536
- 16 bits Identifier– ID of the packet
 - Used for identifying all fragments of the same packet when it is fragmented
 - Flag
 - Fragmentation offset – offset of the first byte of the fragment in its original packet

IP header (3)

- TTL, 8 bits – Time to live
 - Maximum number of hops (router) the packet is allowed to travel
 - Max: 255
 - Router decreases TTL 1 unit when processing a packet
 - The packet will be destroyed when TTL reaches to 0
- Protocol – upper layer protocol
 - Transport protocol (TCP, UDP,...)
 - Other network layer protocols that are encapsulated in IP packet (ICMP, IGMP, OSPF)

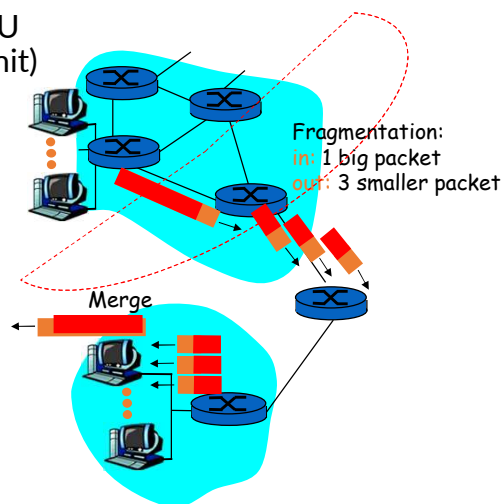
IP header (4)

- Checksum: to detect corruption in the header of IPv4 data packets
- Source IP address
 - 32 bit, address of the sender
- Destination IP address
 - 32 bit, address of the receiver.

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Packet fragmentation (1)

- Each link has a fixed MTU (Maximum transferring unit)
- Different media have different MTU
- If IP packet > MTU, it should be
 - Divided into small fragments
 - Gathered at the destination



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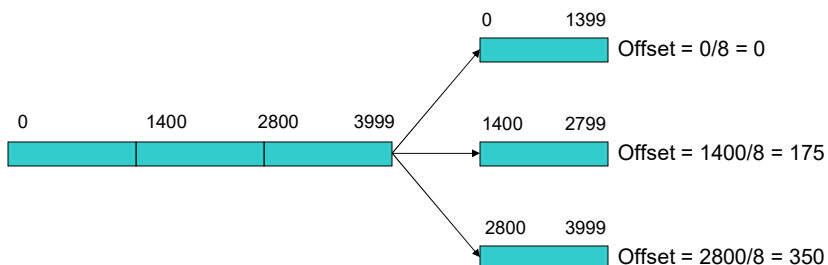
Packet fragmentation (2)

- Identification
 - ID is used to identify fragmentations of the same packet
- Flags – (3 bits)
 - 1st bit: reverse
 - 2nd bit: No more fragments
 - 3rd bit: Still fragments
- Offset
 - Position of the fragments in the original packet
 - In units of 8 bytes

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Packet fragmentation (3)

- Offset
 - Position of the fragment in the original packet
 - In 8 bytes units



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Packet fragmentation (4)

Example

- ❖ Packet: 4020 byte (header: 20 byte)
- ❖ MTU = 1500 bytes

1480 bytes in
data field

offset =
 $1480/8$

length	ID	fragflag	offset
=4020	=x	=0	=0

length	ID	fragflag	offset
=1500	=x	=1	=0

length	ID	fragflag	offset
=1500	=x	=1	=185

length	ID	fragflag	offset
=1060	=x	=0	=370



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4.4. Forwarding IP packet

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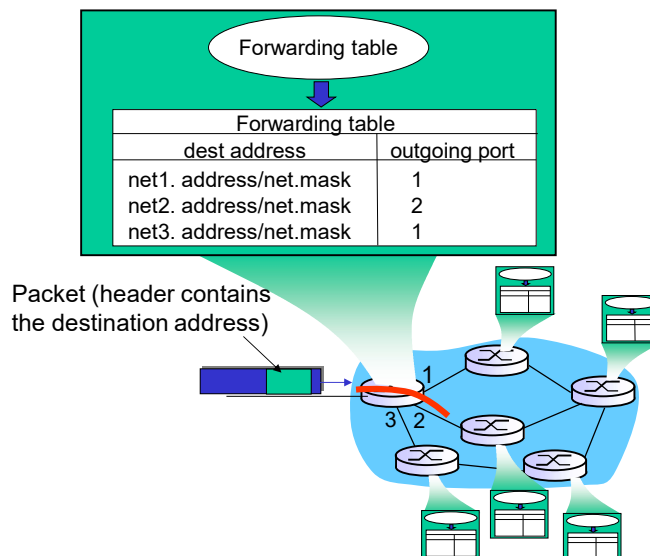
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Chuyển tiếp gói tin IP

- Mỗi nút mạng sử dụng bảng chuyển tiếp (Forwarding Table)
 - Là một phần của bảng định tuyến (Routing Table)
- Các thông tin:
 - Đích đến (Destination): Địa chỉ mạng/Mặt nạ (/n)
 - Lỗi ra mặc định: Sử dụng địa chỉ 0.0.0.0/0 đại diện cho một đích bất kỳ chưa biết
 - Cổng ra (Outgoing port): địa chỉ của cổng ra trên router để chuyển tới nút kế tiếp trong đường đi

Forwarding table



Example – Forwarding table on a host

C:\Documents and Settings\tungbt>netstat -r
Route Table

Interface List

0x1 ... Destination MS TCP Loopback interface
0x2 ... Outgoing port Realtek RTL8139 Family Fast Ethernet NIC -

Network	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	192.168.1.1	192.168.1.34	20
127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1	1
192.168.1.0	255.255.255.0	192.168.1.34	192.168.1.34	20
192.168.1.34	255.255.255.255	127.0.0.1	127.0.0.1	20
192.168.1.255	255.255.255.255	192.168.1.34	192.168.1.34	20
224.0.0.0	240.0.0.0	192.168.1.34	192.168.1.34	20
255.255.255.255	255.255.255.255	192.168.1.34	192.168.1.34	1

Default Gateway: 192.168.1.1

Example – Forwarding table on a router (shorten)

Router# show ip route

O 203.238.37.0/24 via 203.178.136.14, FastEthernet0/1
O 203.238.37.96/27 via 203.178.136.26, Serial0/0/0
C 203.238.37.128/27 is directly connected, Serial0/0/0
O 192.68.132.0/24 via 203.178.136.14, FastEthernet0/1
C 203.254.52.0/24 is directly connected, FastEthernet0/1
C 202.171.96.0/24 is directly connected, Serial0/0/1

Routing table and forwarding mechanism (1)

- Compare the first n bits of the destination packet address with the corresponding bits on the destination network address
 - /n: Destination network mask
 - If there is a match with the destination network, forward to the corresponding port.
 - If no matching destination network is found, forward to the default port (if available)

Destination address of a packet:
11.2.4.10

Destination	Outgoing Port
11.0.0.0 /8	Se0/1
11.1.0.0 /16	Se0/2
11.1.2.0/24	Se0/3

“Longest matching” rule (1)

- Assume that there are more than one entry matching with a destination network in routing table.
- Destination address: 11.1.2.5
- What should be chosen as the next hop?

Network	Next hop
11.0.0.0/8	A
11.1.0.0/16	B
11.1.2.0/24	C

“Longest matching” rule (2)

Destination address:

11.1.2.5 = 00001011.00000001.00000010.00000101

Route 1:

11.1.2.0/24 = 00001011.00000001.00000010.00000000

Route 2:

11.1.0.0/16 = 00001011.00000001.00000000.00000000

Route 3:

11.0.0.0/8 = 00001011.00000000.00000000.00000000

Forwarding on a router

- S1: If TTL = 1 (or TTL = 0), discard the packet and report an error. End.
- S2: If TTL > 1, retrieve the destination address (DA) of the packet. Mask the destination address of the packet with the destination network masks.
- S3: Compare the masked result with the corresponding destination network address. If a matching destination network is found, forward to the corresponding outgoing port, and decrement TTL.
- S4: If no matching destination network is found, check the default outgoing port (corresponding to 0.0.0.0/0).
 - If there is a default port: Forward the packet to the default port and decrement TTL.
 - If there is no default port: Discard the packet and report an error.

4.5. Internet Control Message Protocol

Packet format
Ping and Traceroute

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Idea of ICMP (1)

- IP is unreliable, connectionless
 - Lack of supporting and error control mechanism
- ICMP is used in network layer for providing information exchange between sender and receivers
 - Error information: inform that a packet cannot reach a host, a network or a port.
- Also in network layer but is “above” IP
 - ICMP message is encapsulated in IP
- **ICMP message:** Type, Code, with 8 first bytes of the error IP message



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IP header and Protocol field

Ver	HLEN	DS	Total Length	
Identification			Flags	Fragmentation offset
TTL	Protocol	Header Checksum		
Source IP address				
Destination IP address				
Option				

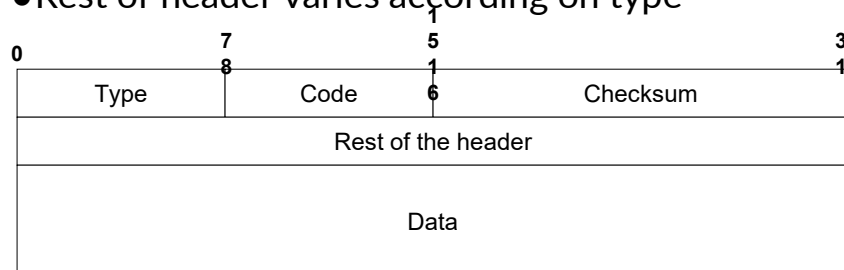
Protocol:

1: ICMP
2: IGMP
6: TCP
17: UDP
89: OSPF

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ICMP message format

- Type: type of ICMP message
- Code: cause of error
- Checksum
- Rest of header varies according on type



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Some ICMP message types

ICMP Message Type	Error-reporting messages	3	Destination Unreachable
		4	Source quench (nguồn giảm tốc độ)
		5	Redirection
		11	Time exceeded
		12	Parameter problem
	Query messages	8 or 0	Echo reply or request
		13 or 14	Time stamp request or reply
		17 or 18	Address mask request or reply
		9 or 10	Router advertisement or solicitation



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ICMP and debugging tools

- ICMP always works transparently for users
- Users can use ICMP by using some debugging tools
 - ping
 - traceroute



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Ping and ICMP

- ping
 - Test a connection
 - Sender sends packet “ICMP echo request”
 - Receiver responds with “ICMP echo reply”
- Data field contains the time stamp when the packet is sent
 - For calculating RTT (round-trip time)

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Ping: Example

```
C:\Documents and Settings\hongson>ping www.yahoo.co.uk
```

Pinging www.euro.yahoo-eu1.akadns.net [217.12.3.11] with 32 bytes of data:

Reply from 217.12.3.11: bytes=32 time=600ms TTL=237

Reply from 217.12.3.11: bytes=32 time=564ms TTL=237

Reply from 217.12.3.11: bytes=32 time=529ms TTL=237

Reply from 217.12.3.11: bytes=32 time=534ms TTL=237

Ping statistics for 217.12.3.11:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

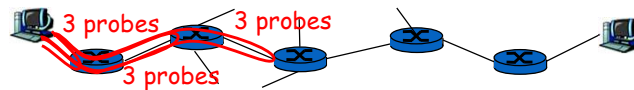
Approximate round trip times in milli-seconds:

Minimum = 529ms, Maximum = 600ms, Average = 556ms

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Traceroute and ICMP

- Syntax:
 - Windows: `tracert host_address`
 - Linux: `traceroute host_address`
- Sender send many packets to receiver
 - First packet has TTL =1
 - Second packet has TTL=2, ...
- When packet number n arrives to nth router:
 - Router destroys the packer
 - Router send back an ICMP packet (type 11, code 0) containing IP address of the router
- Based on the reply message, the sender can calculate RTT



Traceroute and ICMP

Termination condition

- When ICMP echo packet arrive to the destination
- When source receives ICMP “host unreachable” (type 3, code 3)

Traceroute: Example

C:\Documents and Settings\hongson>tracert www.jaist.ac.jp

Tracing route to www.jaist.ac.jp [150.65.5.208]
over a maximum of 30 hops:

```

 1  1 ms  <1 ms  <1 ms  192.168.1.1
 2  15 ms  14 ms  13 ms  210.245.0.42
 3  13 ms  13 ms  13 ms  210.245.0.97
 4  14 ms  13 ms  14 ms  210.245.1.1
 5  207 ms  230 ms  94 ms  pos8-2-br01.hkg04.pccwbtn.net [63.218.115.45]
 6  *      403 ms  393 ms  0.so-0-1-0.XT1.SCL2.ALTER.NET [152.63.57.50]
 7  338 ms  393 ms  370 ms  0.so-7-0-0.XL1.SJC1.ALTER.NET [152.63.55.106]
 8  402 ms  404 ms  329 ms  POS1-0.XR1.SJC1.ALTER.NET [152.63.55.113]
 9  272 ms  288 ms  310 ms  193.ATM7-0.GW3.SJC1.ALTER.NET [152.63.49.29]
10  205 ms  206 ms  204 ms  wide-mae-gw.customer.alter.net [157.130.206.42]
11  427 ms  403 ms  370 ms  ve-13.foundry2.otemachi.wide.ad.jp [192.50.36.62]
12  395 ms  399 ms  417 ms  ve-4.foundry3.nezu.wide.ad.jp [203.178.138.244]
13  355 ms  356 ms  378 ms  ve-3705.cisco2.komatsu.wide.ad.jp [203.178.136.193]
14  388 ms  398 ms  414 ms  c76.jaist.ac.jp [203.178.138.174]
15  438 ms  377 ms  435 ms  www.jaist.ac.jp [150.65.5.208]

```

Trace complete.

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4.6. Network address translation

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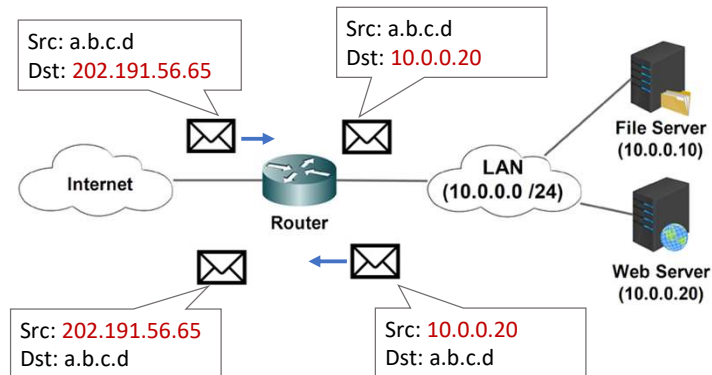
Basic concepts of NAT

- Data transmitted from the LAN (using local addresses) to the Internet (using public addresses) and vice versa needs address conversion
- NAT(Network Address Translation) is performed on the router
- In practice, NAT can be used to convert IP addresses from one LAN to another
- Benefits:
 - Conserves public IP addresses
 - Hides private addresses
 - Reduces configuration costs when changing ISPs

Static NAT

- The simplest form of NAT
- One private IP address is assigned to a specific public IP address
- The NAT router maintains a translation table in memory
 - This table maps internal private addresses to external Internet addresses
- Often used for servers within the LAN to provide services to the public network

Static NAT - Example

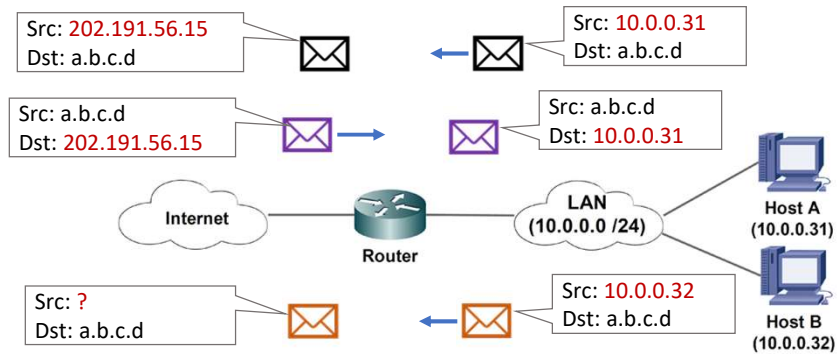


NAT Table	Inside IP	Outside IP
	10.0.0.10	202.191.56.11
	10.0.0.20	202.191.56.65

Dynamic NAT

- The NAT router automatically maps a range of private IP addresses to a range of public IP addresses so that computers can access the Internet when needed.
- In contrast to static NAT, the administrator does not need to configure a specific private IP address to be associated with a specific public IP address.
- Any private IP address will be assigned and automatically translated to one of the available (idle) public IP addresses in the public address pool by the NAT router.

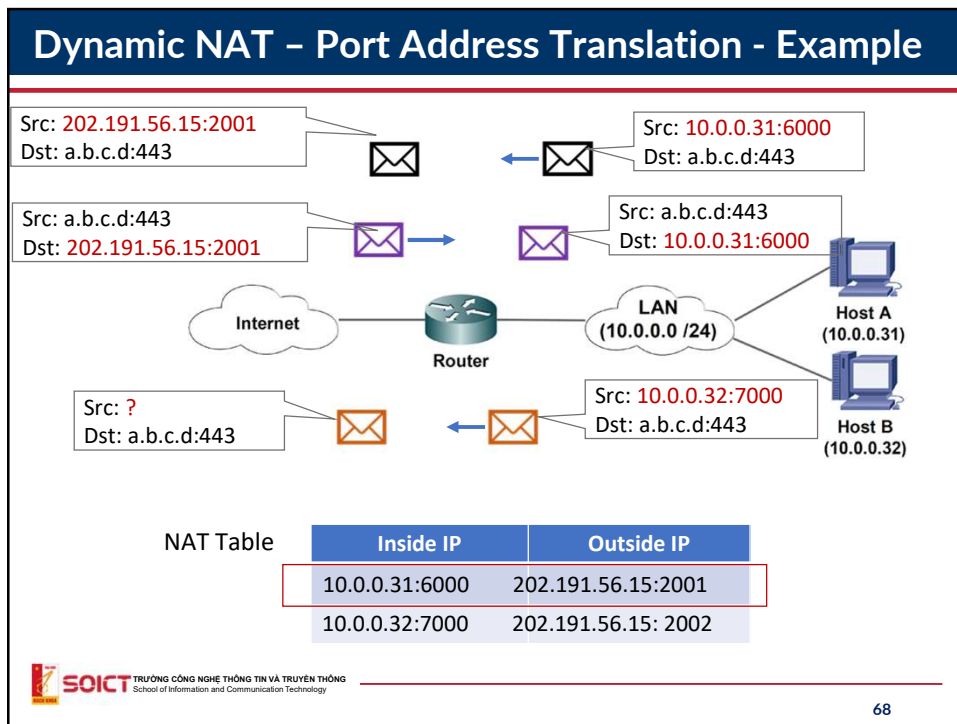
Dynamic NAT – Example



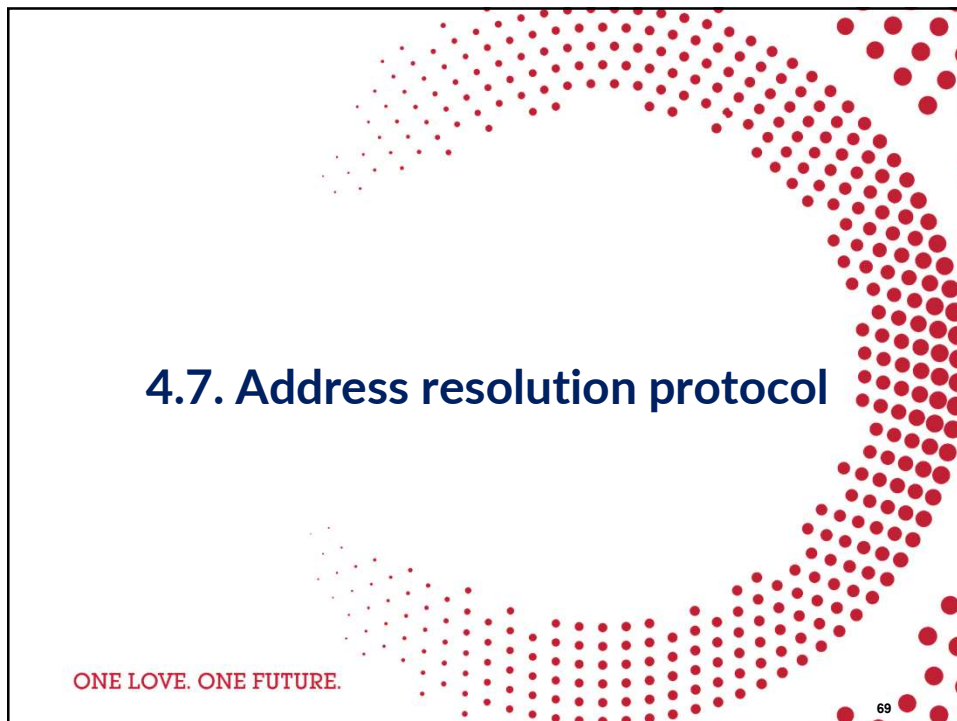
NAT Table	Inside IP	Outside IP
	10.0.0.31	202.191.56.15
	10.0.0.32	202.191.56.16

Port Address Translation (PAT)

- Also known as **NAT overloading**
- It is a special form of dynamic NAT to further utilize the IPv4 address space
- Allows multiple private IP addresses to map to a single registered public IP address with different port numbers
- Thus, PAT uses unique sockets to map to each device



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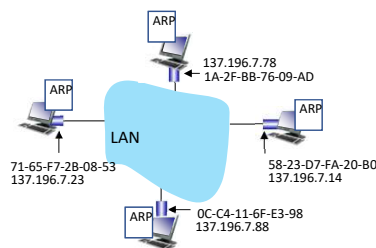
MAC address and ARP

- Address Resolution Protocol
- Finding the MAC address (data link layer identifier) of a network node when the IP address is known
- Why is ARP needed?
 - Communication on the network layer uses IP addresses
 - Communication on the data link layer uses MAC addresses
 - During transmission: data moves from the network layer to the data link layer.
 - Data sent within a LAN: The source device needs to know the MAC address of the destination device
 - Data sent outside the LAN: The source device needs to know the MAC address of the default router

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ARP: address resolution protocol

Question: how to determine interface's MAC address, knowing its IP address?



ARP table: each IP node (host, router) on LAN has table

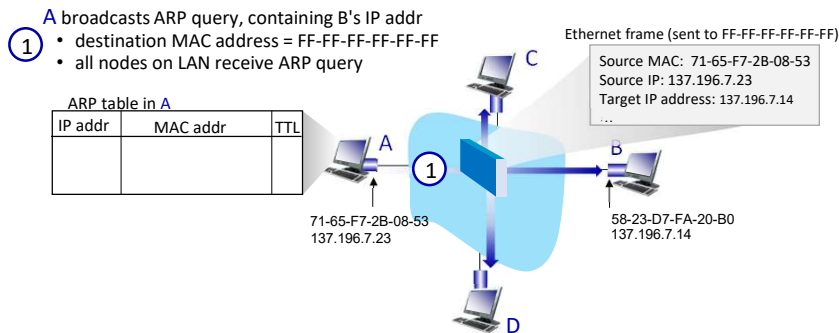
- IP/MAC address mappings for some LAN nodes:
< IP address; MAC address; TTL >
- TTL (Time To Live): time after which address mapping will be forgotten (typically 20 min)

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ARP protocol in action

example: A wants to send datagram to B

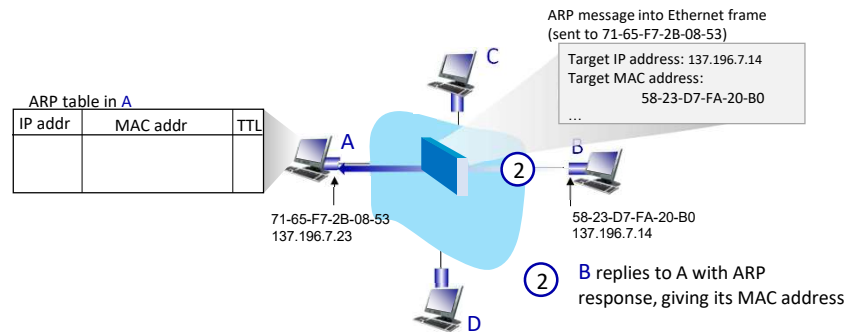
- B's MAC address not in A's ARP table, so A uses ARP to find B's MAC address



ARP protocol in action

example: A wants to send datagram to B

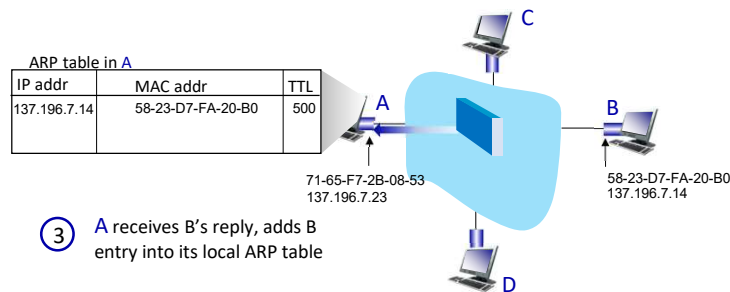
- B's MAC address not in A's ARP table, so A uses ARP to find B's MAC address



ARP protocol in action

example: A wants to send datagram to B

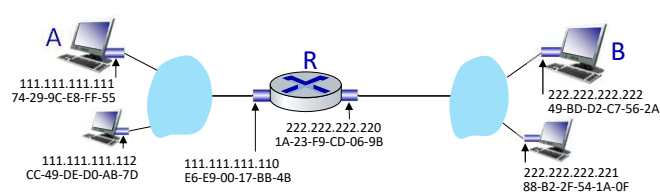
- B's MAC address not in A's ARP table, so A uses ARP to find B's MAC address



Routing to another subnet: addressing

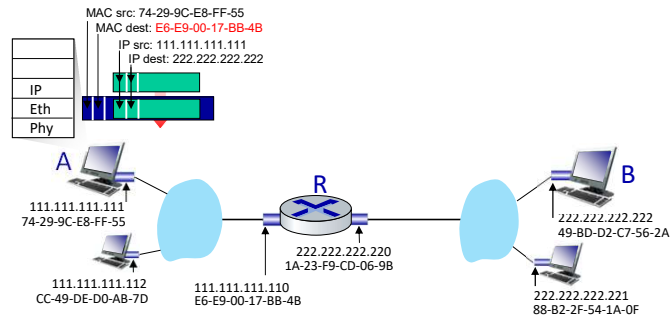
walkthrough: sending a datagram from A to B via R

- focus on addressing – at IP (datagram) and MAC layer (frame) levels
- assume that:
 - A knows B's IP address
 - A knows IP address of first hop router, R (how?)
 - A knows R's MAC address (how?)



Routing to another subnet: addressing

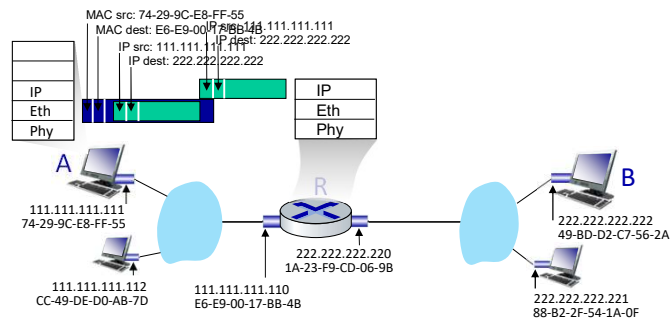
- A creates IP datagram with IP source A, destination B
- A creates link-layer frame containing A-to-B IP datagram
 - R's MAC address is frame's destination



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Routing to another subnet: addressing

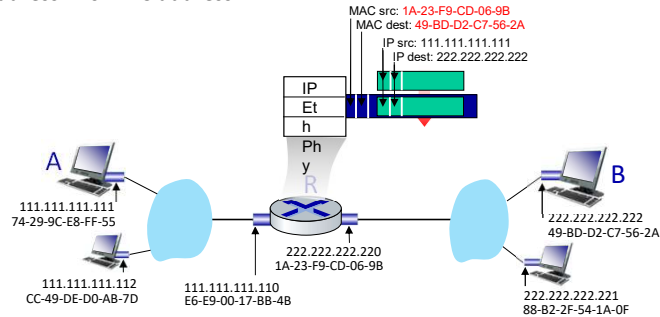
- frame sent from A to R
- frame received at R, datagram removed, passed up to IP



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Routing to another subnet: addressing

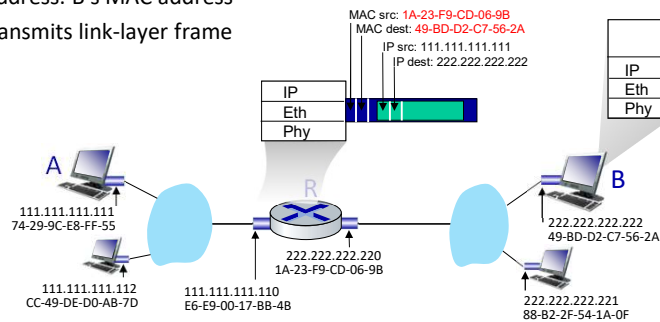
- R determines outgoing interface, passes datagram with IP source A, destination B to link layer
- R creates link-layer frame containing A-to-B IP datagram. Frame destination address: B's MAC address



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Routing to another subnet: addressing

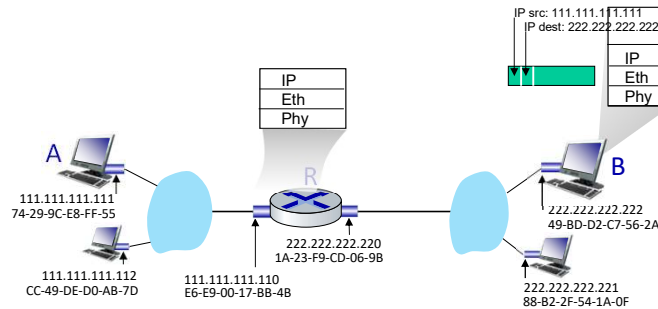
- R determines outgoing interface, passes datagram with IP source A, destination B to link layer
- R creates link-layer frame containing A-to-B IP datagram. Frame destination address: B's MAC address
- transmits link-layer frame



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Routing to another subnet: addressing

- B receives frame, extracts IP datagram destination B
- B passes datagram up protocol stack to IP



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4.8. Dynamic Host Configuration Protocol (DHCP)

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DHCP: Dynamic Host Configuration Protocol

- goal: host dynamically obtains IP address from network server when it “joins” network
 - can renew its lease on address in use
 - allows reuse of addresses (only hold address while connected/on)
 - support for mobile users who join/leave network
- Application layer services provide IP address configuration for network nodes:
 - IP address
 - Subnet mask
 - Default router address (default router, default gateway)
 - May include DNS server address for domain name resolution
- Operates on a client/server model: the client uses IP address parameters assigned by the server

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Các thông điệp DHCP

Client

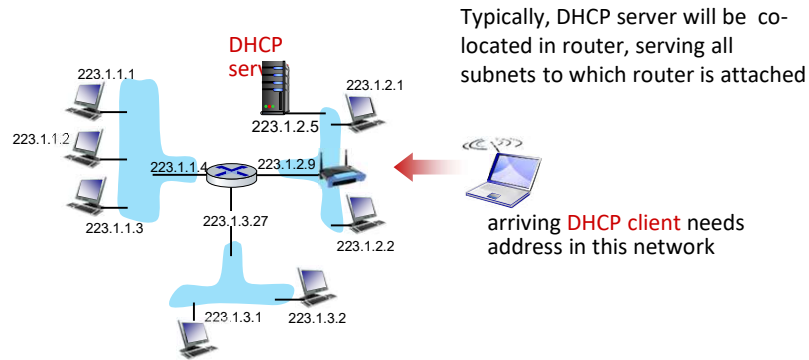
- DHCP Discover: find DHCP Server
- DHCP Request: register to use an IP address
- DHCP Release: release the using IP address
- DHCP Decline: Refuse the offered IP address

Server

- DHCP Offer: Offer the information of IP address
- DHCP ACK: Acknowledge the registration
- DHCP NAK: Refuse the registration

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DHCP client-server scenario

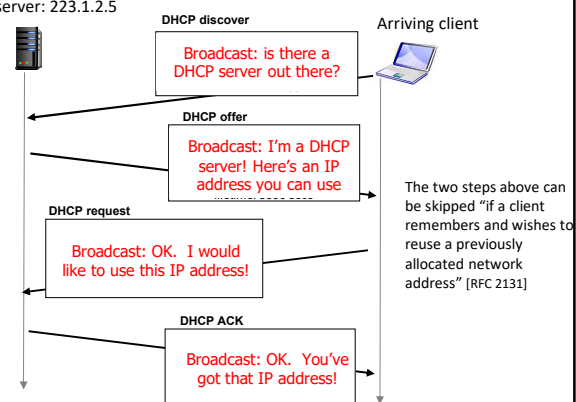


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DHCP client-server scenario

DHCP can return more than just allocated IP address on subnet:

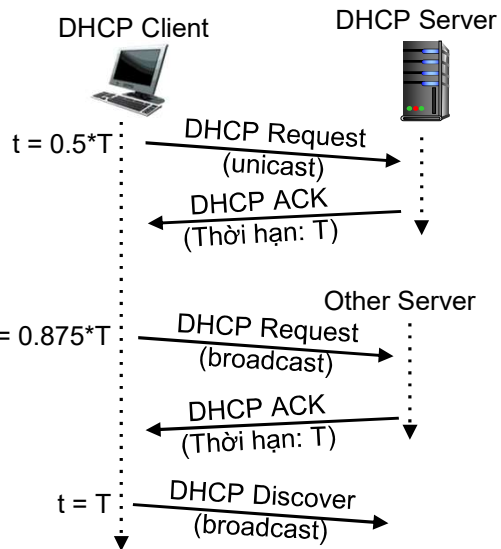
- address of first-hop router for client
- name and IP address of DNS server
- network mask (indicating network versus host portion of address)



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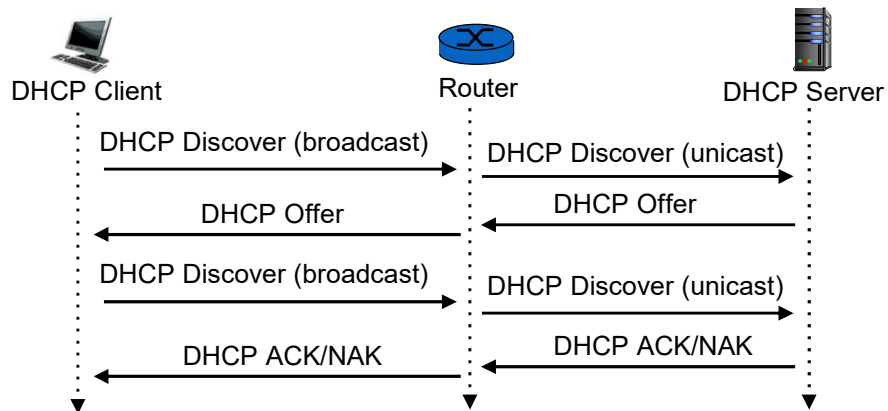
Extend using

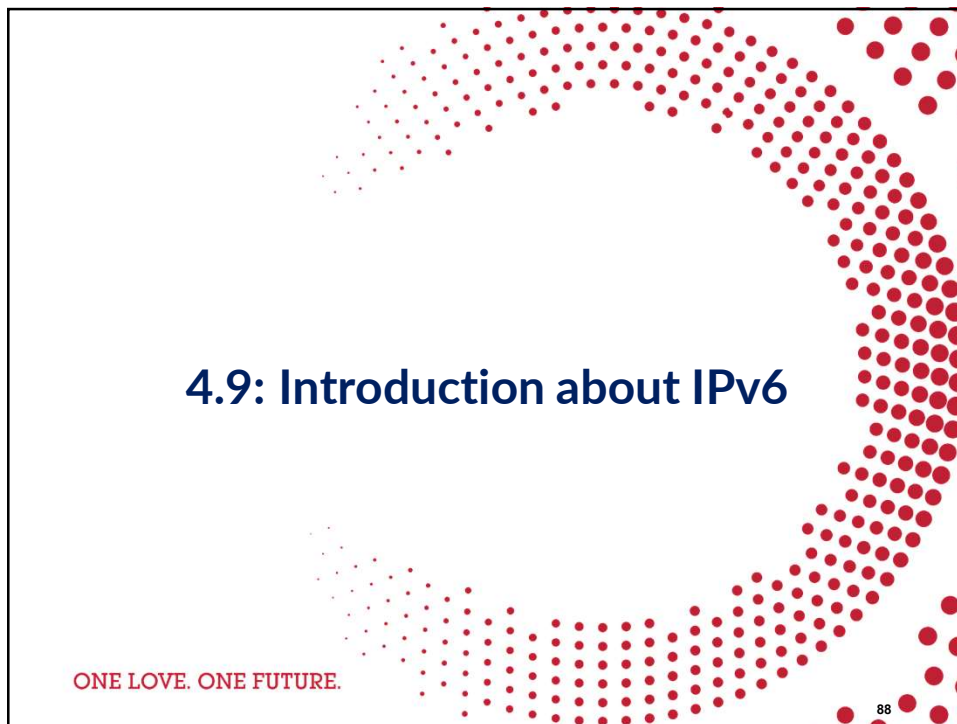
- Limit time $\rightarrow$ extend
- $t = 0.5 \cdot T$, client sends DHCP Request to DHCP Server to request extension
- No DHCP ACK, then $t = 0.875 \cdot T$, client sends the broadcast DHCP Request
- No DHCP ACK, while $t = T$, client sends DHCP Discover



DHCP Relay

- DHCP Server stays on another subnet $\rightarrow$ broadcast packets will be forwarded by routers
- $\rightarrow$ DHCP Relay on routers



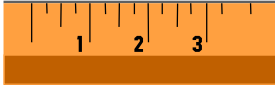


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Giao thức IPv6

- Xuất phát từ nhu cầu thực tế: địa chỉ IPv4 cạn kiệt, không đủ để cấp phát
- Cải tiến trên IPv6:
 - Mở rộng không gian địa chỉ
 - Sử dụng địa chỉ có độ dài 128 bit
 - Phân vùng địa chỉ(scope)

Không gian địa chỉ IPv4



1mm

Không gian địa chỉ IPv6



84.000 lần đường kính của thiên hà



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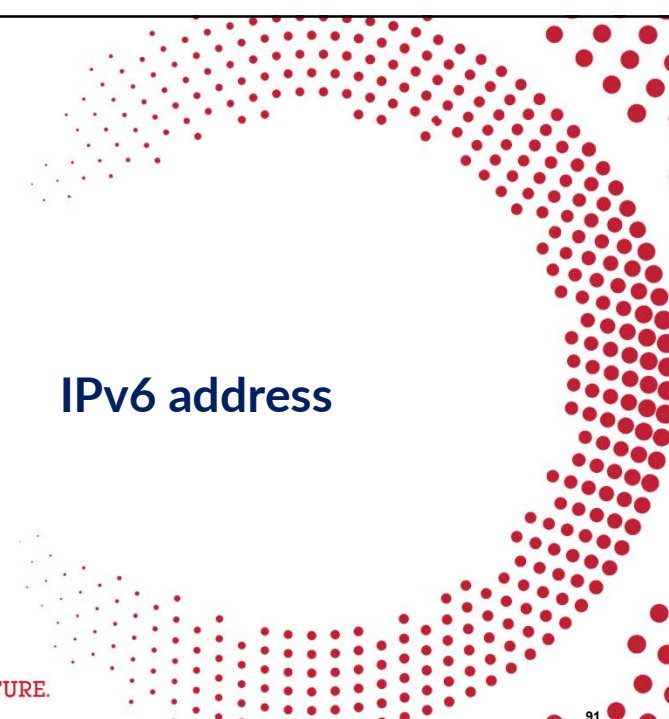
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Cải tiến trên IPv6(tiếp)

- Tăng tốc độ xử lý
 - Khuôn dạng header đơn giản hơn
 - Không phân mảnh gói tin
- Hỗ trợ QoS tốt hơn
- Về an toàn an ninh
 - Sử dụng IPSec như một chuẩn
- Tự động cấu hình
 - Chuẩn hóa cơ chế tự động cấu hình

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IPv6 address

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Representation

- 128 bits, represented in hexadecimal
- Separated by ":" between groups, each consisting of 4 hexadecimal digits
 - Example: 3ffe:501:100c:e320:2e0:18ff:fe98:936d
- Omit consecutive groups of zeros
 - Example: 3ffe:501:100c:e320:0:0:0:1 → 3ffe:501:100c:e320::1
- Using network mask notation /n

Structure of IPv6 addresses

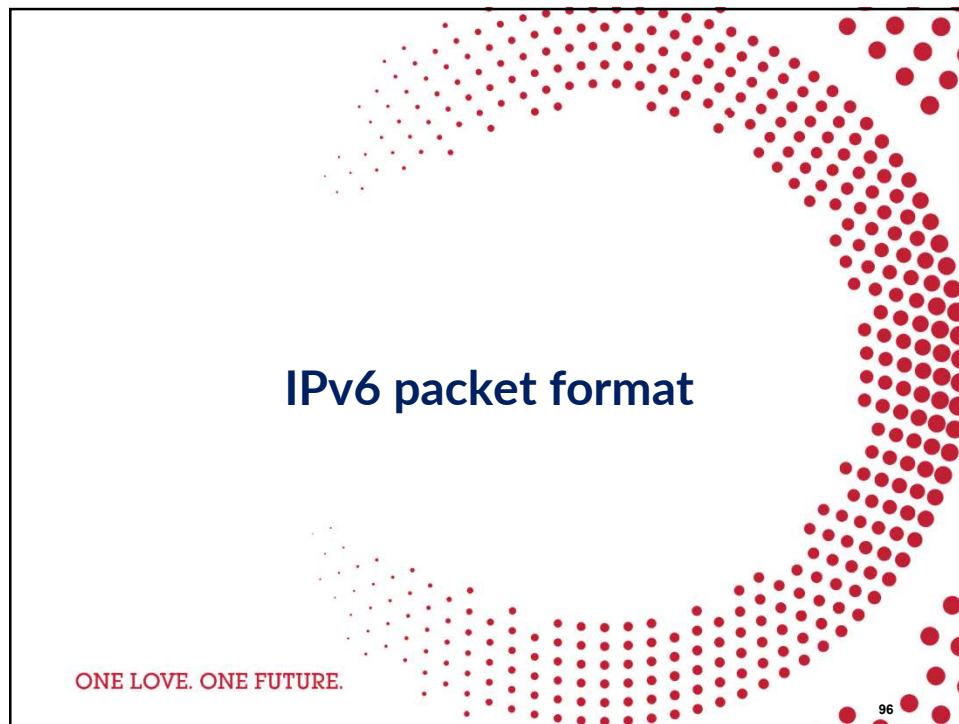
- 2 parts: Network prefix và Interface ID
- Network prefix: 64 bit
- Host ID: 64 bit
 - EUI-64 standard (*extended unique identifier*)
 - In the case of Ethernet networks, the Host ID is determined from the MAC address

Address partitioning

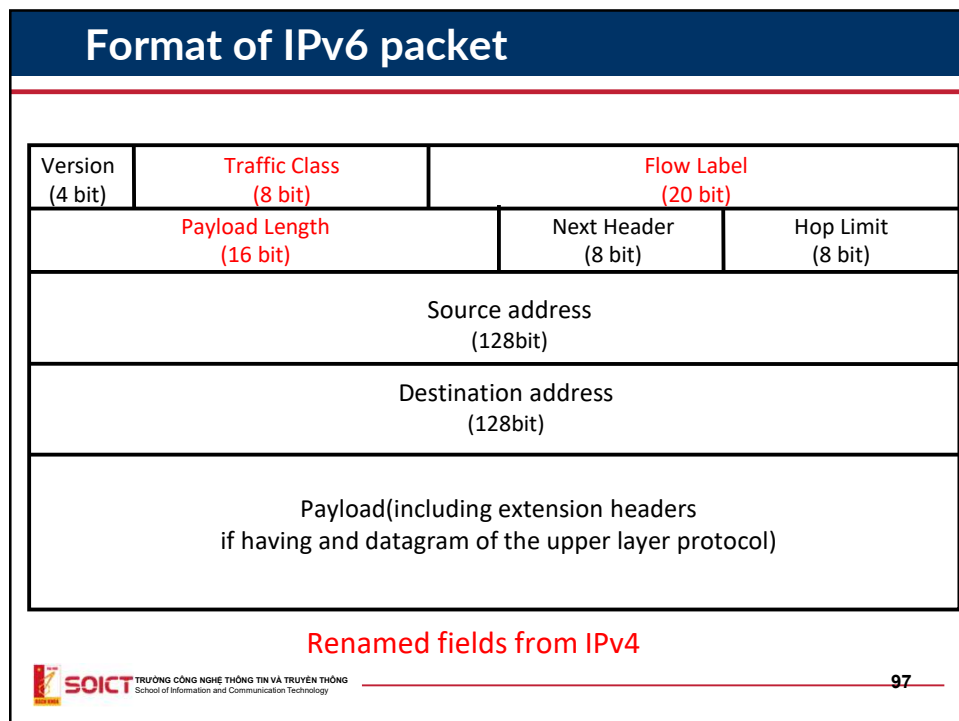
- Global address: Network prefix has
 - Global routing prefix: 48 bits with the first 3 bits being 001
 - SubnetID: 16 bits
- Link-local address: Network prefix has
 - First 10 bits as 1111 1110 10
 - Remaining 54 bits set to 0
- Local address: Network prefix has
 - First 10 bits as 1111 1110 11
 - Next 38 bits set to 0
 - SubnetID: 16 bits

IPv6 address types

- Unicast Address: Assigned to a network interface port.
- Anycast Address: Assigned to a group of network interface ports.
 - Packets sent to an anycast address are forwarded to the nearest node determined by the routing protocol.
- Multicast Address: Assigned to a group of network interface ports within a scope.
 - It begins with 1111 1111.
 - Packets sent to a multicast address are delivered to all nodes in the group.



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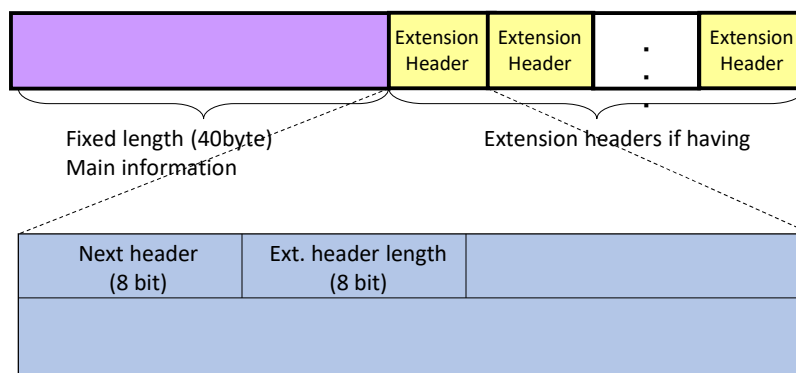
97

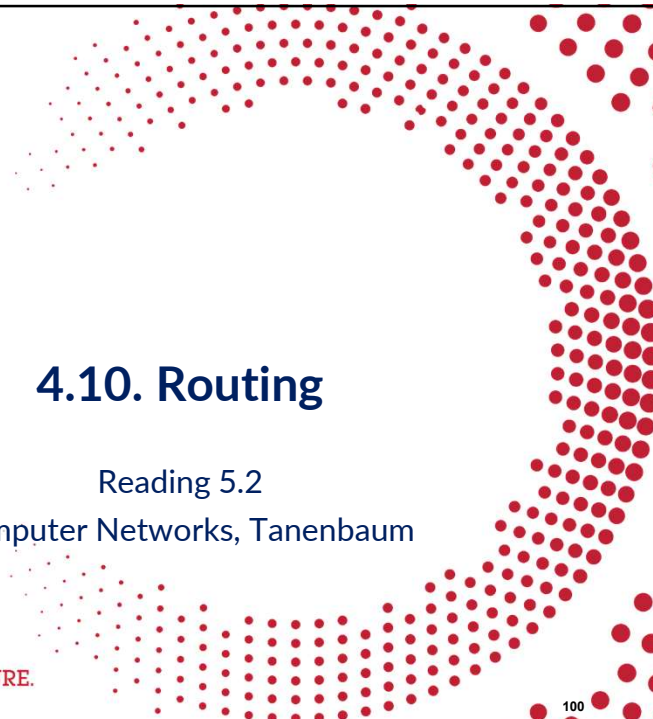
Header of IPv6 packet (1)

- Version: Protocol version (=110)
- Traffic Class: Protocol identifier for the upper layer
- Flow Label: QoS control
- Payload Length: Size of the data portion
- Next header: Next header extension
- Hop limit: Similar to TTL

Header of IPv6 packet (2)

- In IPv6, the extension header is put in the payload section





4.10. Routing

Reading 5.2
Computer Networks, Tanenbaum

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Contents

- What is routing?
- Static routing and dynamic routing
- Routing algorithms and protocols



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What is routing?

Routing principals
Forwarding mechanism
"Longest matching" rule

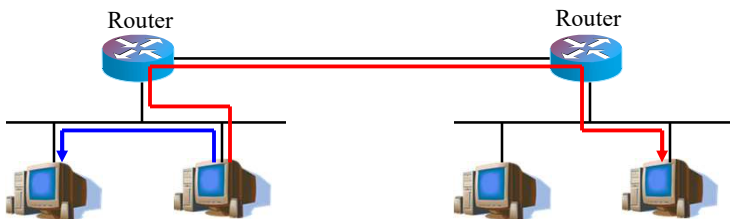
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Routing principles (1)

- When a host send an IP packet to another host
 - If the destination and the source are in the same physical medium: Transfer directly
 - If the destination is in a different network with the source: Send through some other routers (need to choose route)



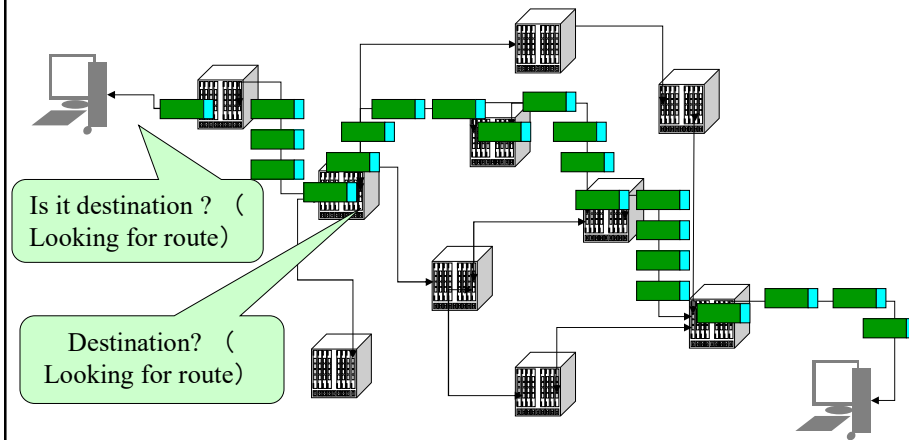
Router Router

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Routing principles (2)



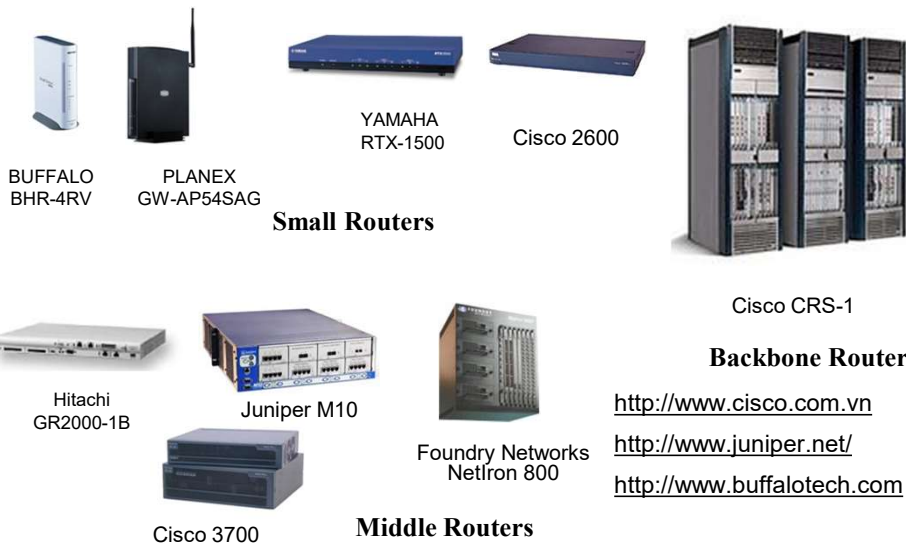
What is routing?

- A mechanism so that a host or a router decides how to forward a packet from source to destination.
- Result of the routing is a routing table
- What to consider in routing
 - Building routing table
 - Information need to calculating route
 - Routing algorithm and protocol.

What is a router?

- Router is the device that forwards data between networks
 - Is a computer with particular hardware
 - Connects multiple networks together, has multiple network interfaces
 - Forward packets according to routing table

Some examples of routers...



Routing table

- Lists of possible routes, saved in the memory of router
- Main components of routing table
 - Destination network address/network mask
 - Next router

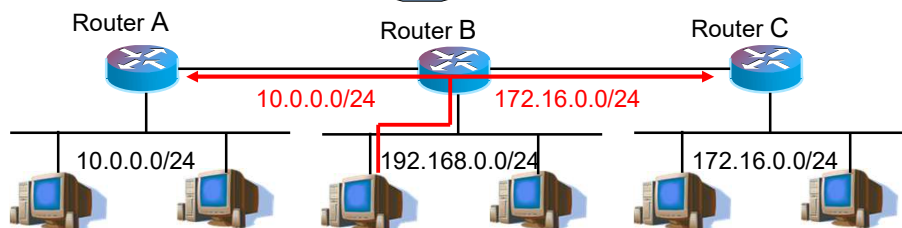
```
#show ip route
```

```
Prefix                Next Hop
203.238.37.0/24 via 203.178.136.14
203.238.37.96/27 via 203.178.136.26
203.238.37.128/27 via 203.178.136.26
203.170.97.0/24 via 203.178.136.14
192.68.132.0/24 via 203.178.136.29
203.254.52.0/24 via 203.178.136.14
202.171.96.0/24 via 203.178.136.14
```

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Routing table and forwarding mechanism (1)

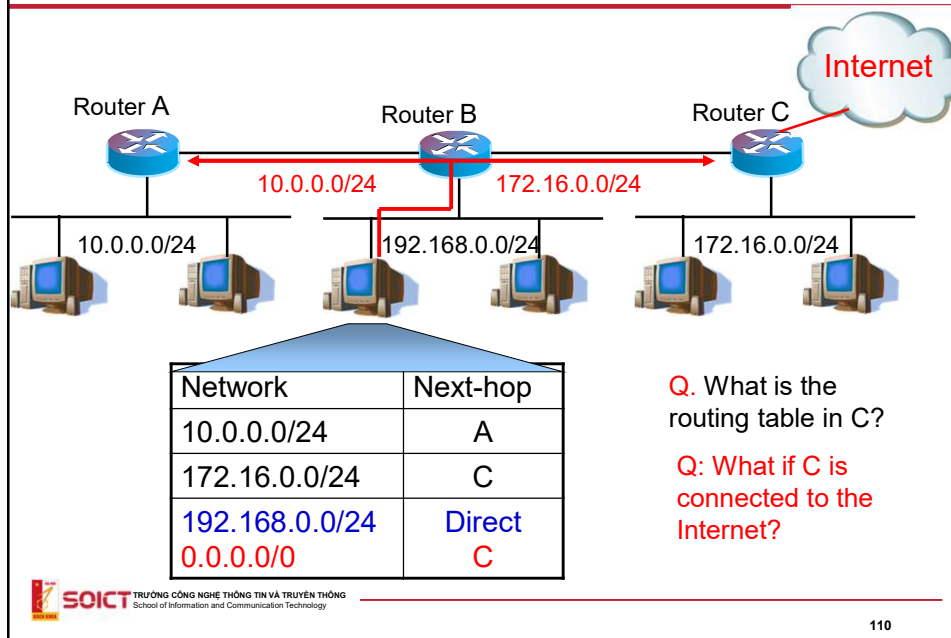
Network	Next-hop
10.0.0.0/24	A
172.16.0.0/24	C



Rule: No routes, no reachability!

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Routing table and forwarding mechanism (2)

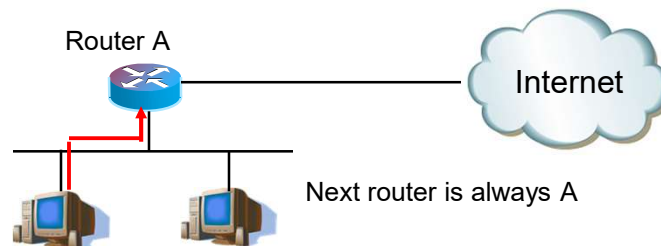


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Default route

- If router does not find a route to a destination in its routing table, default route is necessary
 - Default route is defined for all destination networks that are not figured in the routing table.
- 0.0.0.0/0
 - Is a special notation for all destination networks

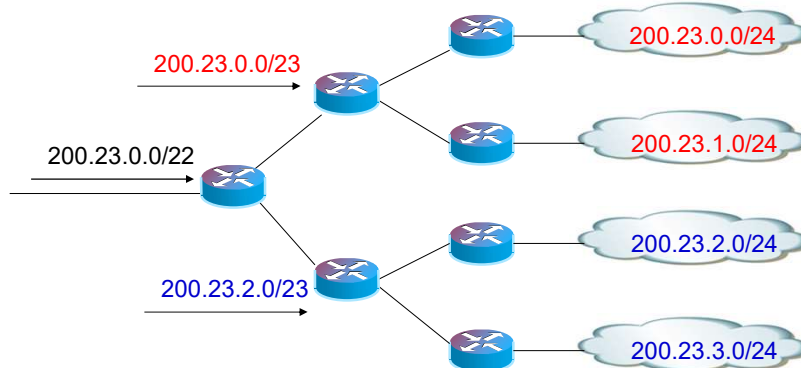


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Route aggregation

- How many networks in the Internet?
- There will be a lot of entries in the routing table?
- The entries to sub-networks of the same “big” network can be aggregated in order to reduce the size of routing table.



Route aggregation (2)

- Example of Viettel network
 - Viettel own a big IP address space
 - 203.113.128.0-203.113.191.255
 - For connecting to a subnet (client) of Viettel, routing table needs only to have a route to Viettel network.
- Default route is a type of route aggregation
 - 0.0.0.0/0

Example – Forwarding table on a host

C:\Documents and Settings\tungbt>netstat -r
Route Table

Interface List

0x1MS TCP Loopback interface
0x2 ...08 00 1f b2 a1 a3 Realtek RTL8139 Family PCI Fast Ethernet NIC -

Network	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	192.168.1.1	192.168.1.34	20
127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1	1
192.168.1.0	255.255.255.0	192.168.1.34	192.168.1.34	20
192.168.1.34	255.255.255.255	127.0.0.1	127.0.0.1	20
192.168.1.255	255.255.255.255	192.168.1.34	192.168.1.34	20
224.0.0.0	240.0.0.0	192.168.1.34	192.168.1.34	20
255.255.255.255	255.255.255.255	192.168.1.34	192.168.1.34	1

Default Gateway: 192.168.1.1

Example – Forwarding table on a router

Router# **show ip route**

O 203.238.37.0/24 via 203.178.136.14, FastEthernet0/1
O 203.238.37.96/27 via 203.178.136.26, Serial0/0/0
C 203.238.37.128/27 is directly connected, Serial0/0/0
O 192.68.132.0/24 via 203.178.136.14, FastEthernet0/1
C 203.254.52.0/24 is directly connected, FastEthernet0/1
C 202.171.96.0/24 is directly connected, Serial0/0/1

Exercises

- A router has the following (CIDR) entries in its routing table:

Address/mask	Next hop	
135.46.56.0/22	Interface 0	0011 1000
135.46.60.0/22	Interface 1	0011 1100
192.53.40.0/23	Router 1	0010 1000
default	Router 2	

- For each of the following IP addresses, what does the router do if a packet with that address arrives?

- (a) 135.46.63.10 0011 1111
 (b) 135.46.57.14
 (c) 135.46.52.2
 (d) 192.53.40.7
 (e) 192.53.56.7 0011 1000

Solution:

Apply longest matching rule.



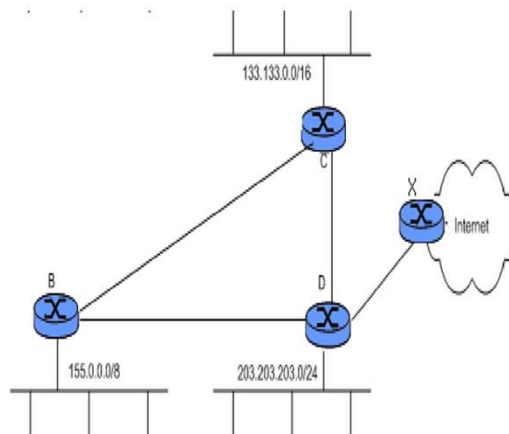
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Exercise

- Assume that we have a network with following topology. What should be routing table of routers B, C, D in order to assure that all hosts can send data to each other and to the Internet.



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Static and dynamic routing

Static routing
Dynamic routing
Advantage - Weakness

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Updating routing table

- Network structure may change
 - New network is added.
 - Router failure due to power corruption ...
- It's necessary to update routing table
 - of all nodes (theory)
 - in practice: some nodes

Network	Next-hop
192.168.0.0/24	B
172.16.0.0/24	B

Network	Next-hop
10.0.0.0/24	A
172.16.0.0/24	C

Network	Next-hop
10.0.0.0/24	B
192.168.0.0/24	B

172.16.1.0/24 B 172.16.1.0/24 C

Router A Router B Router C New Network

10.0.0.0/24 192.168.0.0/24 172.16.0.0/24 172.16.1.0/24

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How to update routing table?

- Static routing
 - Entries in routing table are added manually by administrator
- Dynamic routing
 - Automatically update routing table
 - By mean of routing protocols

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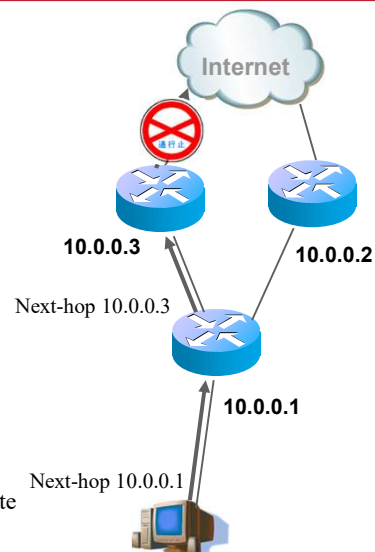
Static routing

- When there is a failure:
 - Problem happens even there are alternative routes.
 - Network administrator needs to change setting

Routing table of 10.0.0.1 (extract)

Prefix	Next-hop
0.0.0.0/0	10.0.0.3

Unreachable route



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Dynamic Routing

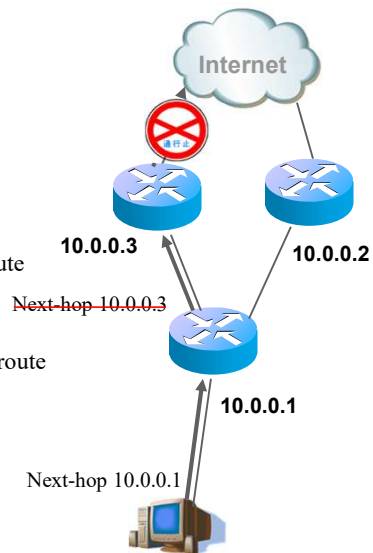
- When there is a failure:
 - Alternative route is added automatically

Routing table of 10.0.0.1 (extract)

Prefix	Next-hop
0.0.0.0/0	10.0.0.3

Alternative route

Unreachable route



Static routing

- Pros
 - Stable
 - Secure
 - It won't be effected by other factors
- Cons
 - Very stubborn
 - Back up link cannot be used
 - Difficult to manage

Dynamic routing

- Pros
 - Easy to manage
 - Backup link can be utilized
- Cons
 - Insecure
 - Difficult to understand the routing protocols

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4.12: Routing algorithm and protocols

Dijkstra and Bellman-Ford Algo
link-state and distance-vector protocols

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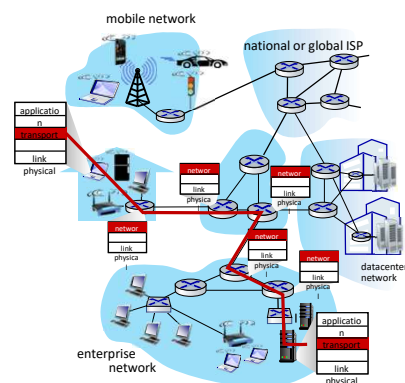
Roles of routing algorithms and protocols

- It is necessary to autonomously construct a routing table at routers.
- Routing algorithms perform path calculations.
- During or before the calculation process, it is essential to gather topology information or exchange information between network nodes.
 - → Routing protocols facilitate this exchange.
 - Routing protocols also define procedures for maintaining the routing table.

Routing protocols

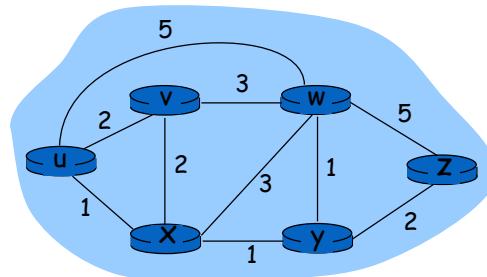
Routing protocol goal:
determine “good” paths (equivalently, routes), from sending hosts to receiving host, through network of routers

- **path:** sequence of routers packets traverse from given initial source host to final destination host
- **“good”:** least “cost”, “fastest”, “least congested”
- routing: a “top-10” networking challenge!



Network as a graph

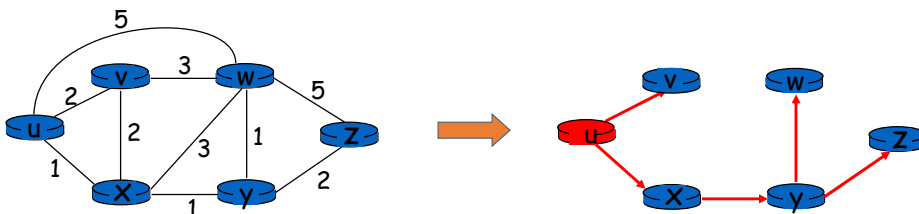
- Graph with nodes (routers) and edges (links)
- Link “cost” $c(x,y)$
 - Bandwidth, delay, cost, congestion level...
- Determine least cost path from every node to every other node



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Shortest path tree - SPT

- A tree that links go out from root to leaves
- The unique path from root to any node v is the shortest path from the root to v
- Each node has a different SPT



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Global or decentralized

- **Routing information** is needed for each node to build routing tables
- Global or decentralized **routing information**?
 - **Global:**
 - all routers have complete topology, link cost info
 - “link state” algorithms
 - **Decentralized:**
 - router knows physically-connected neighbors, link costs to neighbors
 - iterative process of computation, exchange of info with neighbors
 - “distance vector” algorithms
 - “friend of friend is friend”

Dijkstra's link-state routing algorithm

- **centralized:** network topology, link costs known to *all* nodes
 - accomplished via “link state broadcast”
 - all nodes have same info
- computes least cost paths from one node (“source”) to all other nodes
 - gives *forwarding table* for that node
- **iterative:** after k iterations, know least cost path to k destinations

notation

- $C_{x,y}$: direct link cost from node x to y ; $= \infty$ if not direct neighbors
- $D(v)$: *current* estimate of cost of least-cost-path from source to destination v
- $p(v)$: predecessor node along path from source to v
- N' : set of nodes whose least-cost-path *definitively* known

Dijkstra's link-state routing algorithm

```

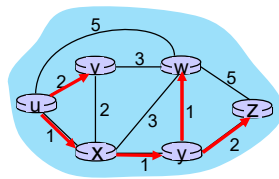
1 Initialization:
2  $N' = \{u\}$  /* compute least cost path from u to all other nodes */
3 for all nodes v
4   if v adjacent to u /* u initially knows direct-path-cost only to direct neighbors */
5     then  $D(v) = c_{u,v}$  /* but may not be minimum cost! */
6   else  $D(v) = \infty$ 
7
8 Loop
9   find w not in  $N'$  such that  $D(w)$  is a minimum
10  add w to  $N'$ 
11  update  $D(v)$  for all v adjacent to w and not in  $N'$  :
12     $D(v) = \min ( D(v), D(w) + c_{w,v} )$ 
13    /* new least-path-cost to v is either old least-cost-path to v or known
14       least-cost-path to w plus direct-cost from w to v */
15 until all nodes in  $N'$ 

```

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Dijkstra's algorithm: an example

Step	N'	$D(v), p(v)$	$D(w), p(w)$	$D(x), p(x)$	$D(y), p(y)$	$D(z), p(z)$
0	u	2, u	5, u	1, u	∞	∞
1	ux	2, u	4, x	∞	2, x	∞
2	uxy	2, u	3, y	∞	∞	4, y
3	uxyv	∞	3, y	∞	∞	4, y
4	uxyvw	∞	∞	∞	∞	4, y
5	uxyvwz	∞	∞	∞	∞	∞

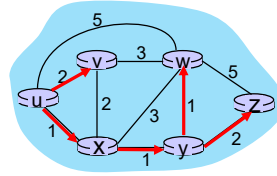


Initialization (step 0): For all a: if a adjacent to then
 $D(a) = c_{u,a}$

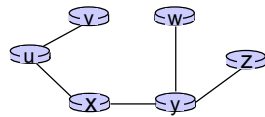
find a not in N' such that $D(a)$ is a minimum
 add a to N'
 update $D(b)$ for all b adjacent to a and not in N' :
 $D(b) = \min (D(b), D(a) + c_{a,b})$

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Dijkstra's algorithm: an example



resulting least-cost-path tree from u:



resulting forwarding table in u:

destination	outgoing link
v	(u,v)
x	(u,x)
y	(u,x)
w	(u,x)
z	(u,x)

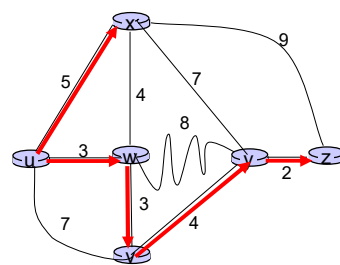
route from u to v directly

route from u to all other destinations via x

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Dijkstra's algorithm: another example

Step	N'	$D(v), p(v)$	$D(w), p(w)$	$D(x), p(x)$	$D(y), p(y)$	$D(z), p(z)$
0	u	7, u	3, u	5, u	∞	∞
1	uw	6, w	5, u	11, w	∞	∞
2	uwvx	6, w		11, w	14, x	
3	uwxv			10, v	14, x	
4	uwxvy				12, y	
5	uwxvyz					



notes:

- construct least-cost-path tree by tracing predecessor nodes
- ties can exist (can be broken arbitrarily)

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Distance vector algorithm

Based on *Bellman-Ford* (BF) equation (dynamic programming):

Bellman-Ford equation

Let $D_x(y)$: cost of least-cost path from x to y .

Then:

$$D_x(y) = \min_v \{ c_{x,v} + D_v(y) \}$$

$\min$ taken over all neighbors v of x

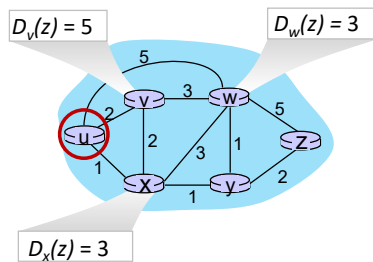
direct cost of link from x to v

v 's estimated least-cost-path cost to y

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Bellman-Ford Example

Suppose that u 's neighboring nodes, x, v, w , know that for destination z :



Bellman-Ford equation says:

$$\begin{aligned} D_u(z) &= \min \{ c_{u,v} + D_v(z), \\ &\quad c_{u,x} + D_x(z), \\ &\quad c_{u,w} + D_w(z) \} \\ &= \min \{ 2 + 5, \\ &\quad 1 + 3, \\ &\quad 5 + 3 \} = 4 \end{aligned}$$

node achieving minimum (x) is
next hop on estimated least-
cost path to destination (z)

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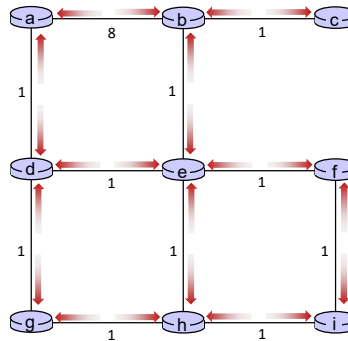
Distance vector example: iteration



t=1

All nodes:

- receive distance vectors from neighbors
- compute their new local distance vector
- send their new local distance vector to neighbors



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Network Layer: 5-142

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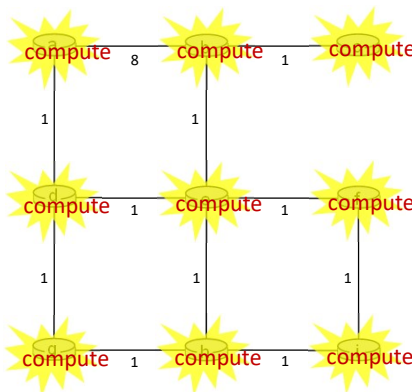
Distance vector example: iteration



t=1

All nodes:

- receive distance vectors from neighbors
- compute their new local distance vector
- send their new local distance vector to neighbors



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Network Layer: 5-143

143

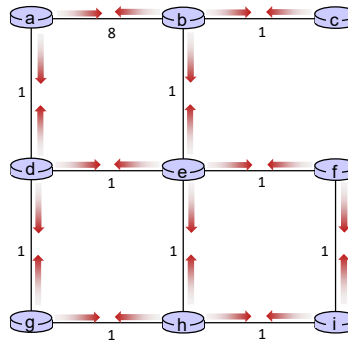
Distance vector example: iteration



t=1

All nodes:

- receive distance vectors from neighbors
- compute their new local distance vector
- send their new local distance vector to neighbors



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Network Layer: 5-144

144

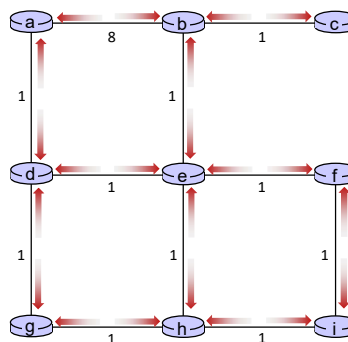
Distance vector example: iteration



t=2

All nodes:

- receive distance vectors from neighbors
- compute their new local distance vector
- send their new local distance vector to neighbors



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Network Layer: 5-145

145

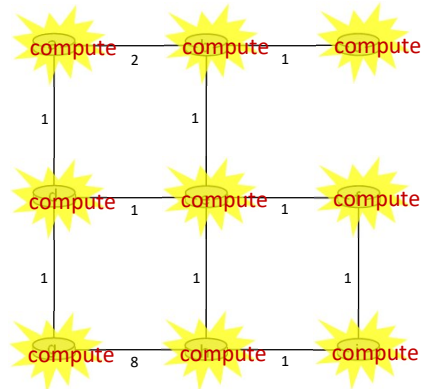
Distance vector example: iteration



t=2

All nodes:

- receive distance vectors from neighbors
- compute their new local distance vector
- send their new local distance vector to neighbors



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Network Layer: 5-146

146

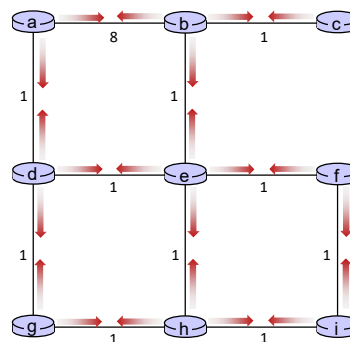
Distance vector example: iteration



t=2

All nodes:

- receive distance vectors from neighbors
- compute their new local distance vector
- send their new local distance vector to neighbors



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Network Layer: 5-147

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Distance vector example: iteration

.... and so on

Let's next take a look at the iterative *computations* at nodes

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Distance vector example: computation



$t=1$

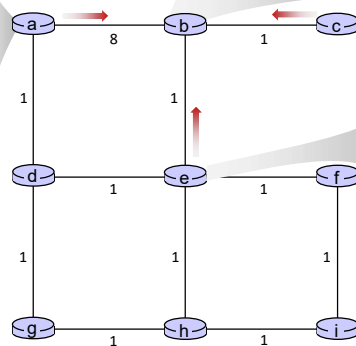
- b receives DVs from a, c, e

DV in a:
$D_a(a) = 0$
$D_a(b) = 8$
$D_a(c) = \infty$
$D_a(d) = 1$
$D_a(e) = \infty$
$D_a(f) = \infty$
$D_a(g) = \infty$
$D_a(h) = \infty$
$D_a(i) = \infty$

DV in b:
$D_b(a) = 8$
$D_b(c) = 1$
$D_b(d) = \infty$
$D_b(e) = 1$
$D_b(f) = \infty$
$D_b(g) = \infty$
$D_b(h) = \infty$
$D_b(i) = \infty$

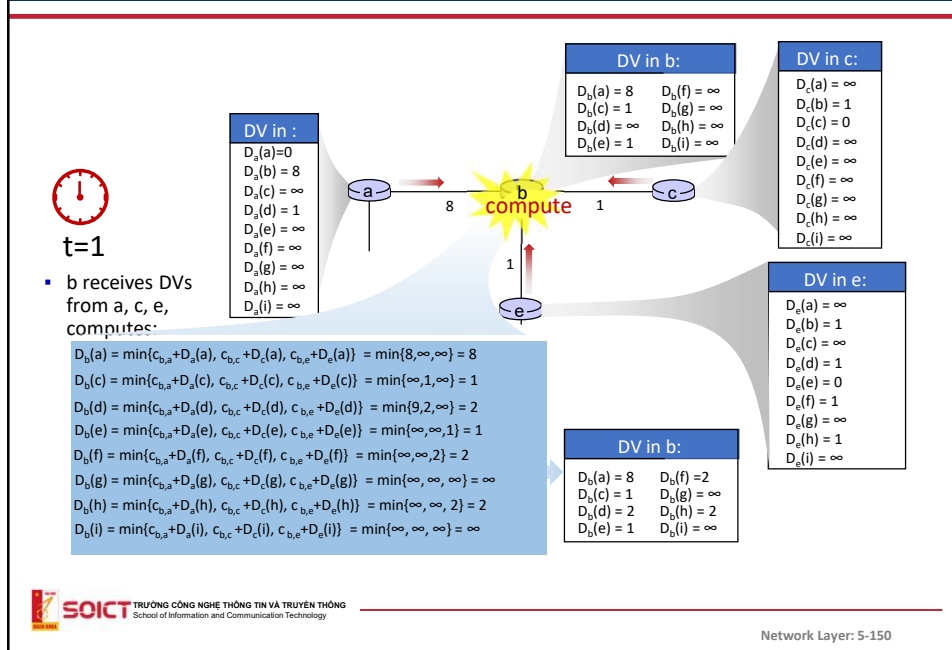
DV in c:
$D_c(a) = \infty$
$D_c(b) = 1$
$D_c(c) = 0$
$D_c(d) = \infty$
$D_c(e) = \infty$
$D_c(f) = \infty$
$D_c(g) = \infty$
$D_c(h) = \infty$
$D_c(i) = \infty$

DV in e:
$D_e(a) = \infty$
$D_e(b) = 1$
$D_e(c) = \infty$
$D_e(d) = 1$
$D_e(e) = 0$
$D_e(f) = 1$
$D_e(g) = \infty$
$D_e(h) = 1$
$D_e(i) = \infty$



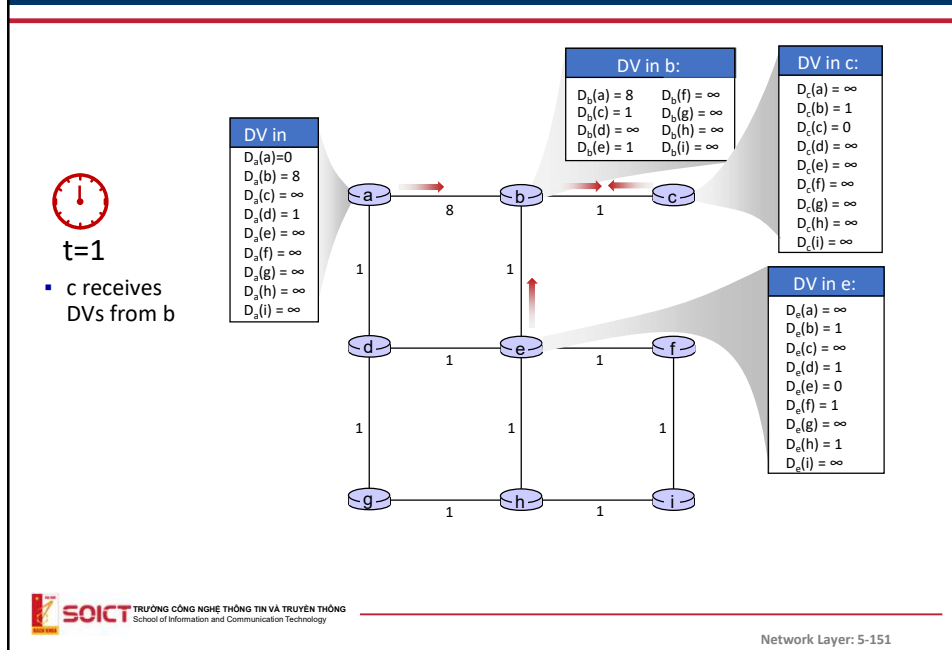
149

Distance vector example: computation



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Distance vector example: computation



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Distance vector example: computation



t=1

- c receives DVs from b computes:

$$\begin{aligned}
 D_c(a) &= \min\{c_{cb} + D_b(a)\} = 1 + 8 = 9 \\
 D_c(b) &= \min\{c_{cb} + D_b(b)\} = 1 + 0 = 1 \\
 D_c(d) &= \min\{c_{cb} + D_b(d)\} = 1 + \infty = \infty \\
 D_c(e) &= \min\{c_{cb} + D_b(e)\} = 1 + 1 = 2 \\
 D_c(f) &= \min\{c_{cb} + D_b(f)\} = 1 + \infty = \infty \\
 D_c(g) &= \min\{c_{cb} + D_b(g)\} = 1 + \infty = \infty \\
 D_c(h) &= \min\{c_{cb} + D_b(h)\} = 1 + \infty = \infty \\
 D_c(i) &= \min\{c_{cb} + D_b(i)\} = 1 + \infty = \infty
 \end{aligned}$$

DV in b:	
$D_b(a) = 8$	$D_b(f) = \infty$
$D_b(c) = 1$	$D_b(g) = \infty$
$D_b(d) = \infty$	$D_b(h) = \infty$
$D_b(e) = 1$	$D_b(i) = \infty$

DV in c:	
$D_c(a) = \infty$	
$D_c(b) = 1$	
$D_c(c) = 0$	
$D_c(d) = \infty$	
$D_c(e) = \infty$	
$D_c(f) = \infty$	
$D_c(g) = \infty$	
$D_c(h) = \infty$	
$D_c(i) = \infty$	

DV in c:	
$D_c(a) = 9$	
$D_c(b) = 1$	
$D_c(c) = 0$	
$D_c(d) = \infty$	
$D_c(e) = 2$	
$D_c(f) = \infty$	
$D_c(g) = \infty$	
$D_c(h) = \infty$	
$D_c(i) = \infty$	

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Distance vector example: computation



t=1

- e receives DVs from b, d, f, h

DV in d:	
$D_d(a) = 1$	
$D_d(b) = \infty$	
$D_d(c) = \infty$	
$D_d(d) = 0$	
$D_d(e) = 1$	
$D_d(f) = \infty$	
$D_d(g) = 1$	
$D_d(h) = \infty$	
$D_d(i) = \infty$	

DV in h:	
$D_h(a) = \infty$	
$D_h(b) = \infty$	
$D_h(c) = \infty$	
$D_h(d) = \infty$	
$D_h(e) = 1$	
$D_h(f) = \infty$	
$D_h(g) = 1$	
$D_h(h) = 0$	
$D_h(i) = 1$	

DV in b:	
$D_b(a) = 8$	$D_b(f) = \infty$
$D_b(c) = 1$	$D_b(g) = \infty$
$D_b(d) = \infty$	$D_b(h) = \infty$
$D_b(e) = 1$	$D_b(i) = \infty$

DV in e:	
$D_e(a) = \infty$	
$D_e(b) = 1$	
$D_e(c) = \infty$	
$D_e(d) = 1$	
$D_e(e) = 0$	
$D_e(f) = 1$	
$D_e(g) = \infty$	
$D_e(h) = 1$	
$D_e(i) = \infty$	

DV in f:	
$D_f(a) = \infty$	
$D_f(b) = \infty$	
$D_f(c) = \infty$	
$D_f(d) = \infty$	
$D_f(e) = 1$	
$D_f(f) = 0$	
$D_f(g) = \infty$	
$D_f(h) = \infty$	
$D_f(i) = 1$	

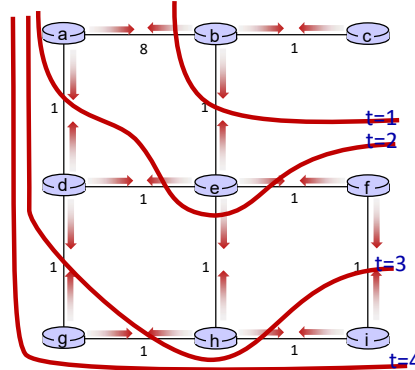
Q: what is new DV computed in e at t=1?

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Distance vector: state information diffusion

Iterative communication, computation steps diffuses information through network:

- 🕒 $t=0$ c's state at $t=0$ is at c only
- 🕒 $t=1$ c's state at $t=0$ has propagated to b, and may influence distance vector computations up to 1 hop away, i.e., at b
- 🕒 $t=2$ c's state at $t=0$ may now influence distance vector computations up to 2 hops away, i.e., at b and now at a, e as well
- 🕒 $t=3$ c's state at $t=0$ may influence distance vector computations up to 3 hops away, i.e., at b,a,e and now at c,f,h as well
- 🕒 $t=4$ c's state at $t=0$ may influence distance vector computations up to 4 hops away, i.e., at b,a,e, c, f, h and now at g,i as well

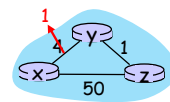


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Distance vector: link cost changes

link cost changes:

- node detects local link cost change
- updates routing info, recalculates local DV
- if DV changes, notify neighbors



"good news travels fast"

t_0 : y detects link-cost change, updates its DV, informs its neighbors.

t_1 : z receives update from y, updates its table, computes new least cost to x, sends its neighbors its DV.

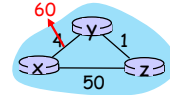
t_2 : y receives z's update, updates its distance table. y's least costs do *not* change, so y does *not* send a message to z.

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Distance vector: link cost changes

link cost changes:

- node detects local link cost change
- “bad news travels slow” – count-to-infinity problem:



- y sees direct link to x has new cost 60, but z has said it has a path at cost of 5. So y computes “my new cost to x will be 6, via z); notifies z of new cost of 6 to x.
- z learns that path to x via y has new cost 6, so z computes “my new cost to x will be 7 via y), notifies y of new cost of 7 to x.
- y learns that path to x via z has new cost 7, so y computes “my new cost to x will be 8 via y), notifies z of new cost of 8 to x.
- z learns that path to x via y has new cost 8, so z computes “my new cost to x will be 9 via y), notifies y of new cost of 9 to x.
- ...
- Distributed algorithms are tricky!*

Comparison of Link-state and Distance vector

Number of exchange messages

- LS: n nodes, E links, $O(nE)$ messages
- DV: Exchange only with neighbor

Convergent time

- LS: Complexity $O(n^2)$
- DV: Varies

Reliability: If some routers provide incorrect information

LS:

- The router may send out incorrect cost
- Each node calculate its own routing table

DV:

- Incorrect distance vector may be sent out
- Each node calculate its DV based to what receives from the neighbor
 - Error propagates in the network.

4.13: Hierarchical routing

Autonomous System
Intra and Inter domain routing

ONE LOVE. ONE FUTURE.

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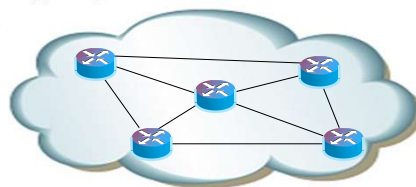
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Making routing scalable

our routing study thus far - id

- all routers identical
- network "flat"

... not true in practice



scale: billions of destinations:

- can't store all destinations in routing tables!
- routing table exchange would swamp links!

administrative autonomy:

- Internet: a network of networks
- each network admin may want to control routing in its own network

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1 Hierarchical structure of the Internet

- Internet = network of networks
- Such networks can select its own routing policy (routing domain)
- Such networks are called autonomous system (AS)

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2 Autonomous System (AS)

- A set of routers with the same routing policy (routing protocol, metric...) is aggregated into an AS
- Gateway: router connect between two ASes
- Each AS has an unique number (ASN - 16 bits or 32 bits).

2914 NTT-COMMUNICATIONS-2914 - NTT America, Inc.
3491 BTN-ASN - Beyond The Network America, Inc.
4134 CHINANET-BACKBONE No.31,Jin-rong Street
6453 GLOBEINTERNET Teleglobe America Inc.
24087 VNGT-AS-AP Vietnam New Generation Telecom
24066 VNNIC-AS-VN Vietnam Internet Network Information Center
17981 CAMBOTECH-KH-AS ISP Cambodia

Source: <http://www.cidr-report.org>

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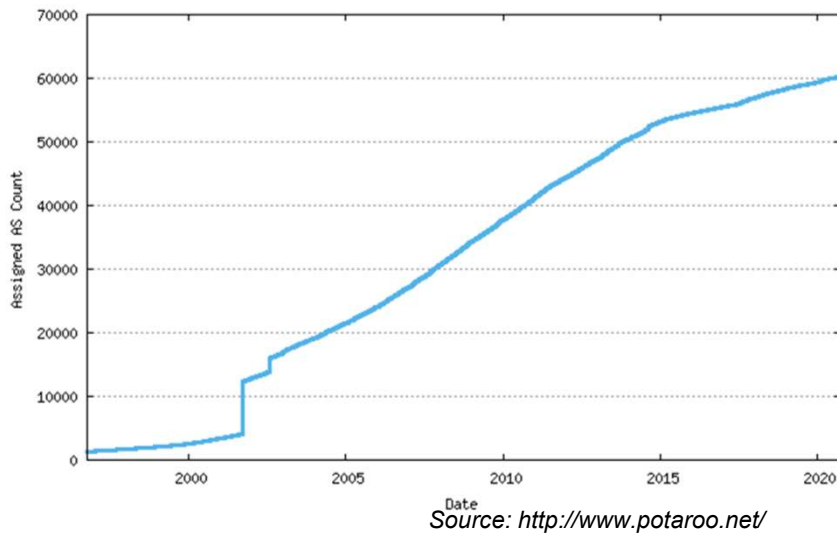
Slide 160

- 1 Combine 5 and 6
sonnh, 3/8/2008

Slide 161

- 2 Explain about AS
sonnh, 3/9/2008

Number of AS by time



Internet approach to scalable routing

aggregate routers into regions known as “autonomous systems” (AS) (a.k.a. “domains”)

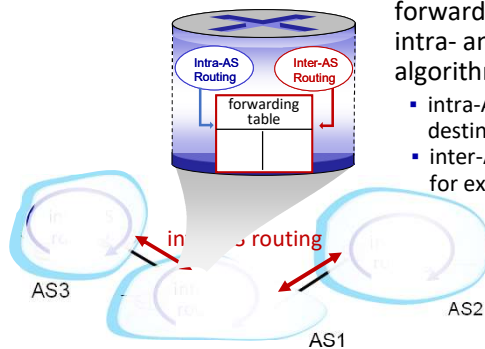
intra-AS (aka “intra-domain”):
routing among *within same AS*
 (“network”)

- all routers in AS must run same intra-domain protocol
- routers in different AS can run different intra-domain routing protocols
- **gateway router:** at “edge” of its own AS, has link(s) to router(s) in other AS'es

inter-AS (aka “inter-domain”):
routing *among* AS'es

- gateways perform inter-domain routing (as well as intra-domain routing)

Interconnected ASes

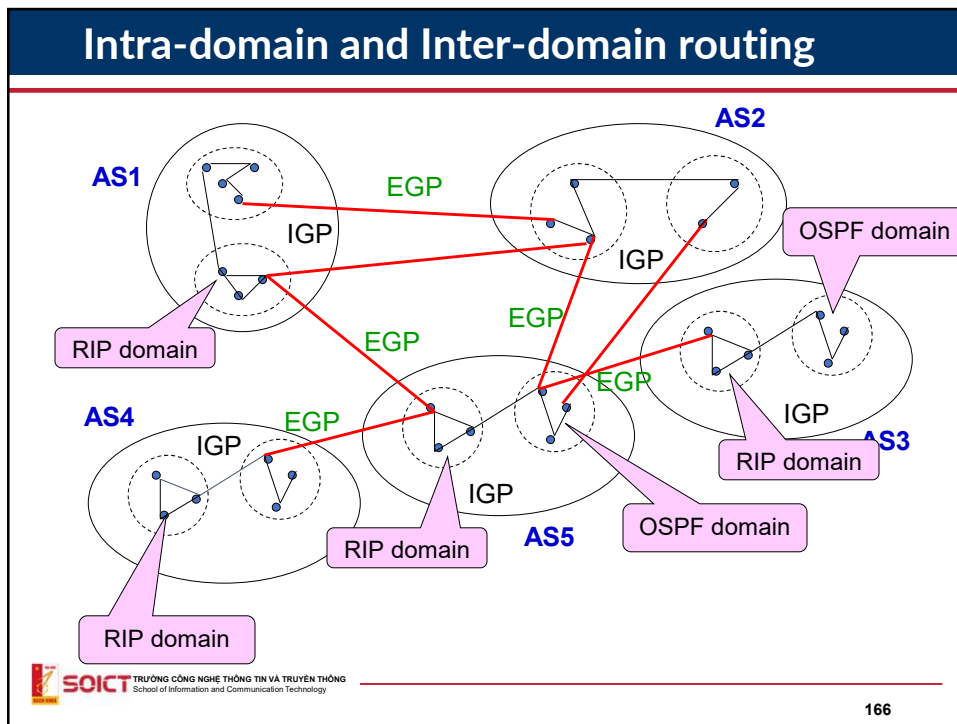


forwarding table configured by intra- and inter-AS routing algorithms

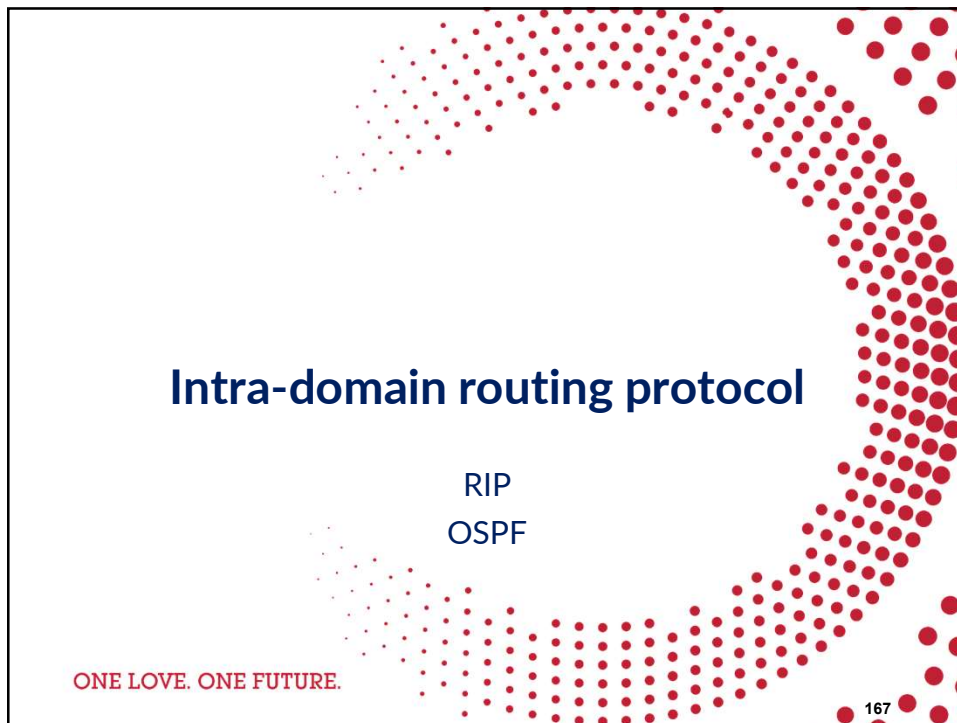
- intra-AS routing determine entries for destinations within AS
- inter-AS & intra-AS determine entries for external destinations

Hierarchical routing protocols

- **Inside an AS:** Intra-domain routing protocols
 - Also named IGP: *Interior Gateway Protocol*
 - RIP: Routing Information Protocol
 - OSPF: Open Shortest Path First
 - IS-IS, IGRP, EIGRP (Cisco)...
- **Among ASes:** Inter-domain routing protocols
 - Also named EGP: *Exterior Gateway Protocol*
 - BGP (v4): Border Gateway Protocol



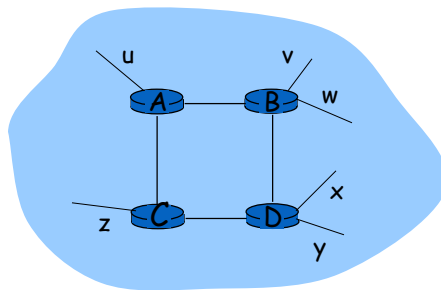
166



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RIP (Routing Information Protocol)

- IGP
- RIP v.1, currently use RIP v.2
- Distance-vector algorithm
- Routing metric: # of hops (max = 15 hops)

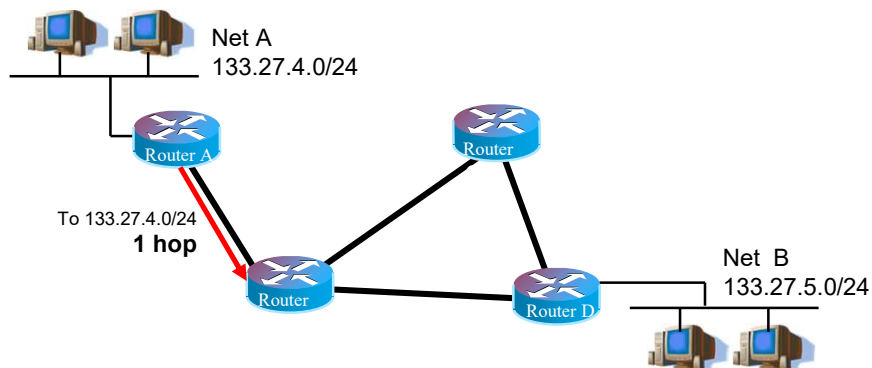


From router A to subsets:

destination	hops
u	1
v	2
w	2
x	3
y	3
z	2

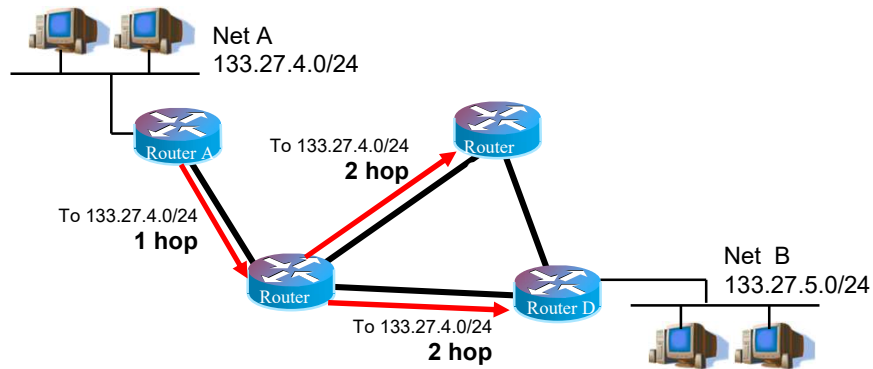
Review: DV routing (1)

- Friend of friend is friend



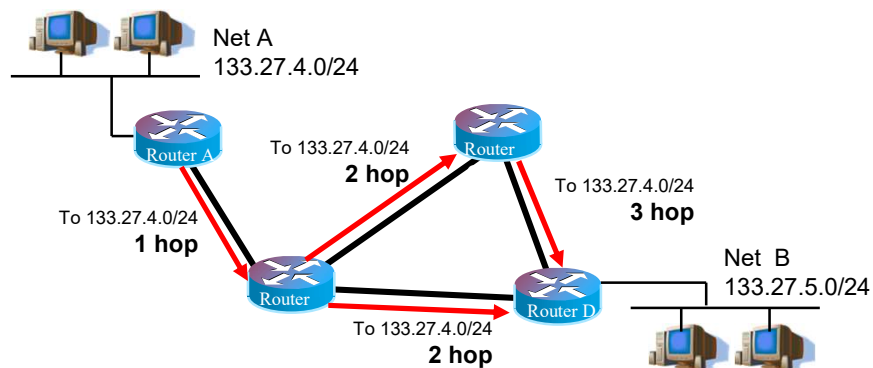
Review: DV routing (2)

- Friend of friend is friend



Review: DV routing (3)

- Friend of friend is friend



4 3 Review: DV routing (4)

- Friend of friend is friend

Net A
133.27.4.0/24

Router A

To 133.27.4.0/24
1 hop

Router

Router D

Net B
133.27.5.0/24

To 133.27.4.0/24
2 hop

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RIP: Exchange information

- Routing table of router is exchanged
- Periodic
 - Node advertise its distance-vector with neighbors every 30s
 - Each message contains up to 25 routing table entries
 - In practice, multiple messages are sent
- Triggered
 - When every entry changes, send copy of entry to neighbors
 - Neighbors use to update their tables

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Slide 172

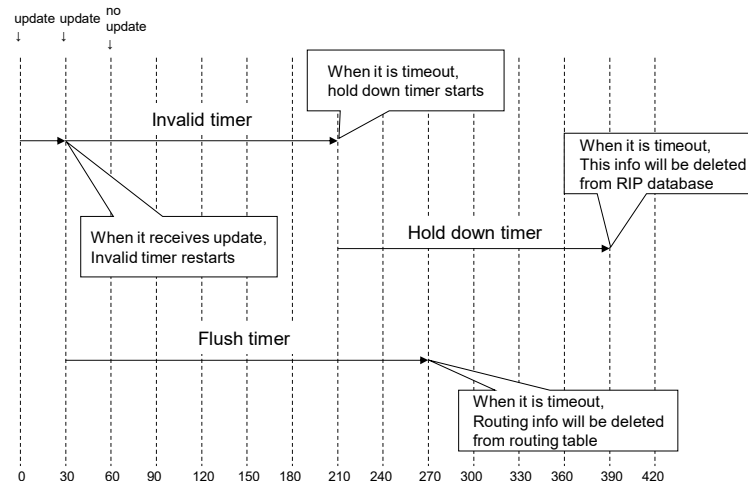
- 4 Explain in opposite way: How B is annouced
sonnh, 3/9/2008
- 3 Expain that we announce networks address. not router id
sonnh, 3/9/2008

RIP timer (1)

- Update timer
 - Exchange routing table every 30 sec
- Invalid timer
 - Updated every time router receives information
 - If it is time out (180sec), it becomes hold down status
- Hold down timer
 - Router keeps routing information for 180 sec
 - Not refer the worse update (to avoid the loop)
 - Possibly down status
- Flush timer
 - Update every time router receives information
 - If 240sec passed, routing entry will be deleted

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RIP timer (2)



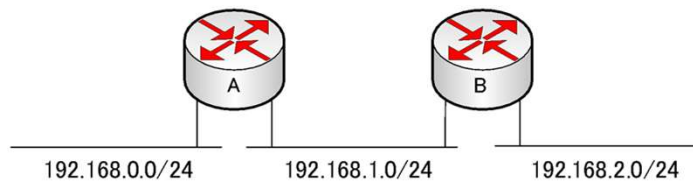
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Ping-pong failure

- If 192.168.0.0/24 is down...
 - B can update 192.168.0.0 info to A
 - Packets to 192.168.0.0/24 become loop status
- A will update 192.168.0.0 info to B
 - Count up to infinity!

192.168.0.0/24	conn
192.168.1.0/24	conn
192.168.2.0/24	B

192.168.1.0/24	conn
192.168.2.0/24	conn
192.168.0.0/24	A



RIP: to avoid this loop

- Limit the maximum hop count
 - 16
- Split horizon
 - The routing info will not return back to the sender
- Poison reverse
 - When network is down, send update with metric 16
 - That routing information become Hold-down status

- 5 16 TTL vs. this?
sonnh, 3/9/2008

OSPF (Open Shortest Path First) routing

- *IGP*
- “open”: publicly available - standard by IETF (current version, version 3, defined in [RFC 2740](#))
- classic link-state
 - each router floods OSPF link-state advertisements (directly over IP rather than using TCP/UDP) to all other routers in entire AS
 - multiple link costs metrics possible: bandwidth, delay
 - each router has full topology, uses Dijkstra’s algorithm to compute forwarding table (*Shortest Path First*)
- *Advanced features*
 - *security*: all OSPF messages authenticated (to prevent malicious intrusion)
 - Large AS: *Hierarchical* OSPF
 - *Classless* routing (able to use Variable-Length Subnet Masking -VLSM)
 - Different *metric* for each link based on TOS (is not used in practice)

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Hierarchical OSPF

- Why we have to divide the network into small area?
- If we have more than 100 routers....
 - Link state update is delivered all the time
 - Number of re-calculation increase
 - Need more memory, need more CPU power
 - Number of link state update become large
 - Routing table become large



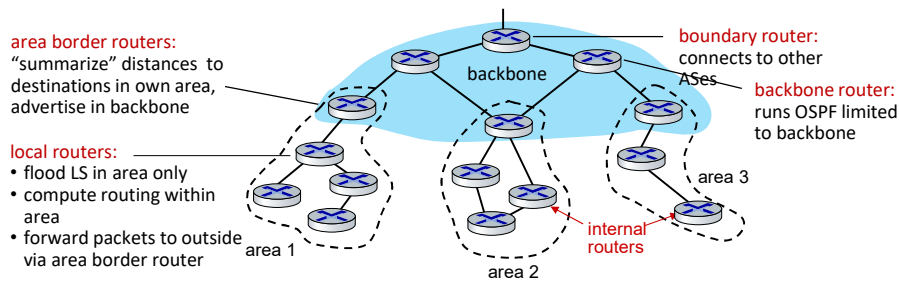
- Area
 - Group of routers which share the same LSA

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- 6 Explain why we need to reduce the calculaiton
sonnh, 3/9/2008

Hierarchical OSPF

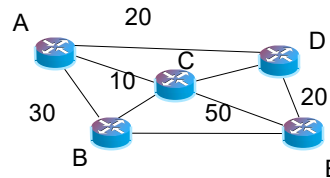
- **two-level hierarchy:** local area, backbone.
 - link-state advertisements flooded only in area, or backbone
 - each node has detailed area topology; only knows direction to reach other destinations



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Which information is exchanged among routers?

- Link-State Advertisement (LSA): Contain the link and the cost to the neighbor
- For example, node A
 - link to B, cost 30
 - link to D, cost 20
 - link to C, cost 10
- For example, node D
 - link to A, cost 20
 - link to E, cost 20
 - link to C, cost 50



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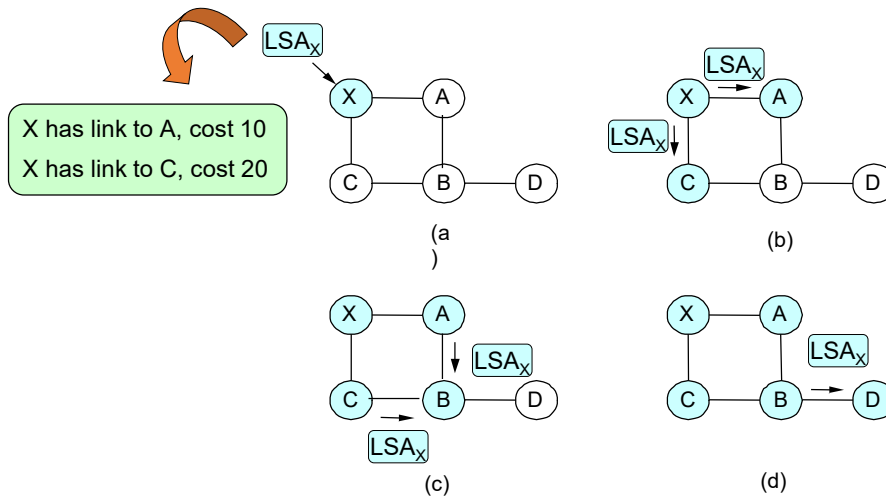
OSPF metric

- Default value is existing
100Mbps / bandwidth of interface
 - But these days administrator assign the original value
- During the calculation of routing table
 - Smallest cost to the one path will be selected
- If cost are same
 - Router will do load balancing

OSPF default cost

Link Bandwidth	Default OSPF cost
56Kbps serial link	1785
64Kbps serial link	1562
T1 (1.544Mbps) serial link	65
E1 (2.048Mbps) serial link	48
4Mbps Token Ring	25
Ethernet	10
16Mbps Token Ring	6
FDDI or Fast Ethernet	1
Gigabit Ethernet / 10G network	1

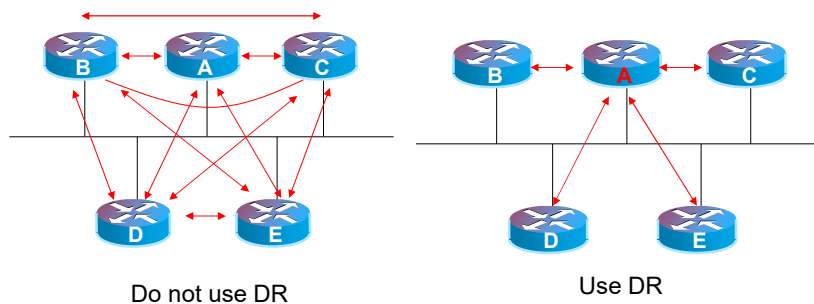
Link state flooding



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Designated router (DR)

- To improving efficiency on exchanging link state
- Each router forms an adjacency with the designated router (DR)
 - Exchanging link state through DR
 - If DR fails, use a BDR (Backup DR)
- How to select DR and BDR?



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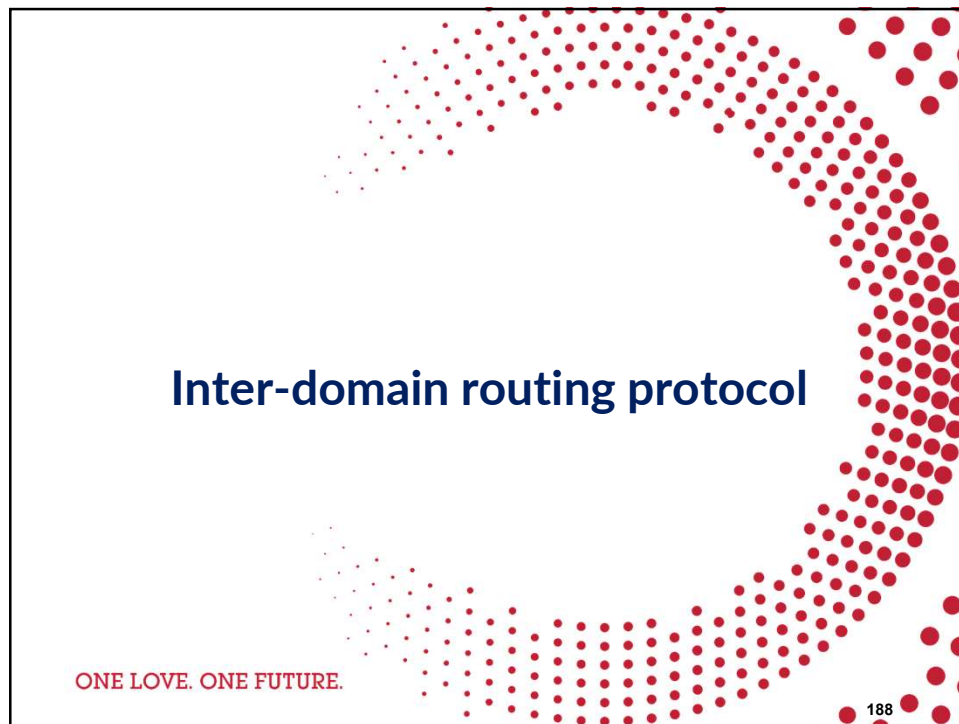
Neighbor & Adjacency

- Neighbor and adjacency are different concept!
 - Adjacency: router which exchange the routing information each other
 - Neighbor: routers have a direct link
- Broadcast multi-access network (e.g Ethernet)
 - Neighbor != Adjacency
- Point-to-point network
 - Neighbor == Adjacency
- Non Broadcast Multi-access network (e.g. ATM)
 - Exchanging data using unicast

RIP vs. OSPF

	RIP	OSPF
Characteristics	• Flat relation between routers • Implementation is simple • Small-scale network	• Support hierarchy • Implementation is complicated • Middle and large-scale network
Scalability	X	O
Computational complexity	little	many
Convergence time	Low speed	high speed
Exchanged information	Routing table	Link-state information
Algorithm	Distant vector type	Link-state type
Neighbor discovery	30s	10s (Hello packet)
Metric	number of hops	Cost (band width)

- 7 Exchanged informatio
Rip:
OSPF
sonnh, 3/9/2008



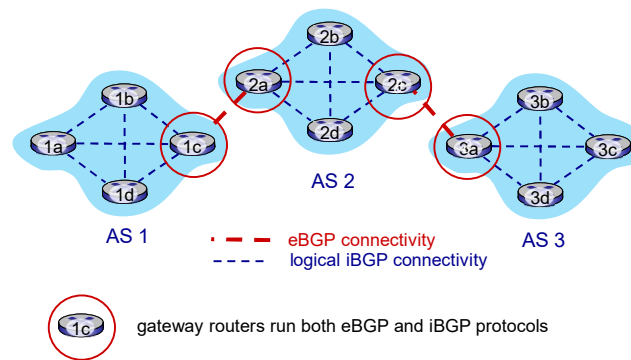
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Internet inter-AS routing: BGP

- **BGP (Border Gateway Protocol):** *the de facto inter-domain routing protocol*
 - “glue that holds the Internet together”
- allows subnet to advertise its existence, and the destinations it can reach, to rest of Internet: *“I am here, here is who I can reach, and how”*
- BGP provides each AS a means to:
 - **eBGP:** obtain subnet reachability information from neighboring ASes
 - **iBGP:** propagate reachability information to all AS-internal routers.
 - determine “good” routes to other networks based on reachability information and *policy*

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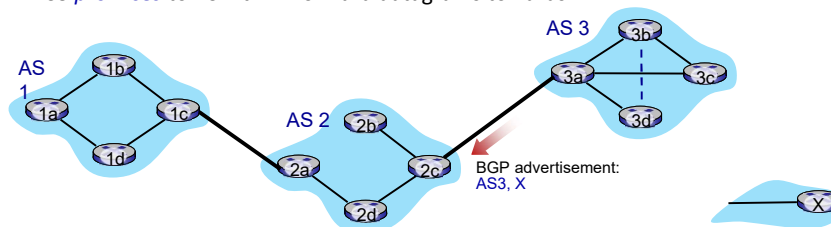
eBGP, iBGP connections



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BGP basics

- **BGP session:** two BGP routers ("peers") exchange BGP messages over semi-permanent TCP connection:
 - advertising *paths* to different destination network prefixes (BGP is a "path vector" protocol)
- when AS3 gateway 3a advertises *path AS3,X* to AS2 gateway 2c:
 - AS3 *promises* to AS2 it will forward datagrams towards X



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8 BGP: Path vector routing

- Which routing protocol can be used to connect multiple ASes?
 - No universal metric – policy decisions
 - LS: No, Metric are not the same, LS database too large – entire Internet
 - DV: Bellman-Ford algorithm may not converge
- Solution: Path vector routing

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Path attributes and BGP routes

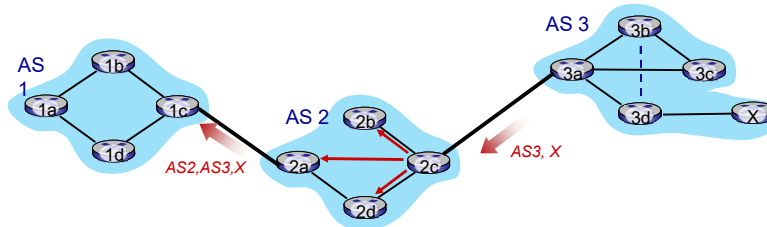
- BGP advertised route: prefix + attributes
 - prefix: destination being advertised
 - two important attributes:
 - AS-PATH**: list of ASes through which prefix advertisement has passed
 - NEXT-HOP**: indicates specific internal-AS router to next-hop AS
- policy-based routing**:
 - gateway receiving route advertisement uses *import policy* to accept/decline path (e.g., never route through AS Y).
 - AS policy also determines whether to *advertise* path to other neighboring ASes

Network Layer: 5-193

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- 8 Change the symbol of router into AS
sonnh, 2/29/2008

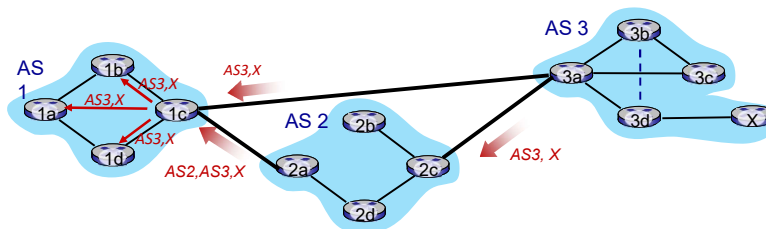
BGP path advertisement



- AS2 router 2c receives path advertisement **AS3,X** (via eBGP) from AS3 router 3a
- based on AS2 policy, AS2 router 2c accepts path AS3,X, propagates (via iBGP) to all AS2 routers
- based on AS2 policy, AS2 router 2a advertises (via eBGP) path **AS2, AS3, X** to AS1 router 1c

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BGP path advertisement (more)



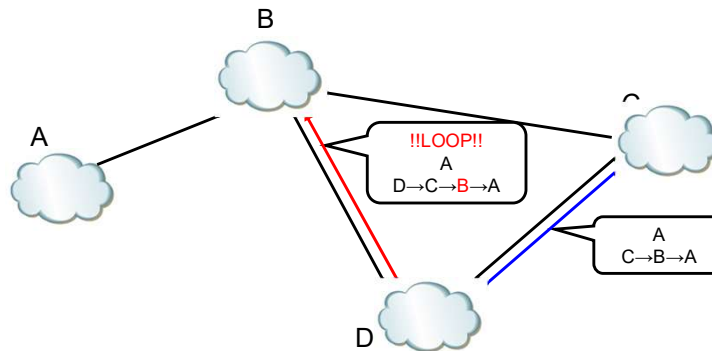
gateway router may learn about **multiple** paths to destination:

- AS1 gateway router 1c learns path **AS2,AS3,X** from 2a
- AS1 gateway router 1c learns path **AS3,X** from 3a
- based on **policy**, AS1 gateway router 1c chooses path **AS3,X** and advertises path within AS1 via iBGP

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Loop free mechanism

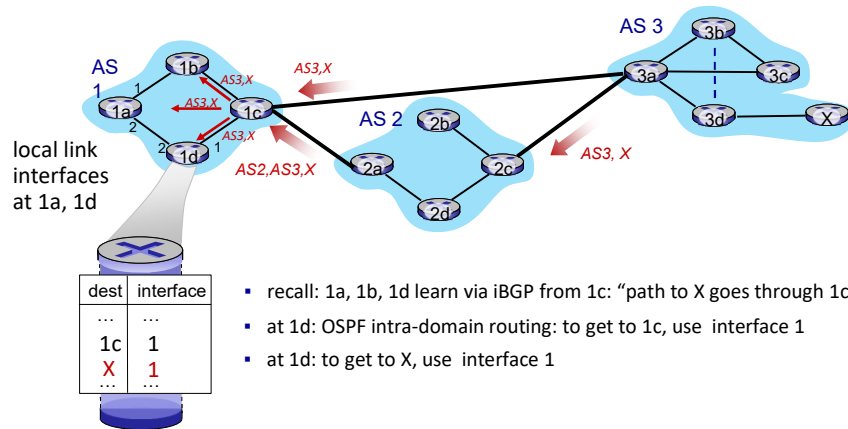
- Detecting loop depending on whether that router is included in path of received routing information or not
 - B will cancel the route to A, which B includes in path



BGP messages

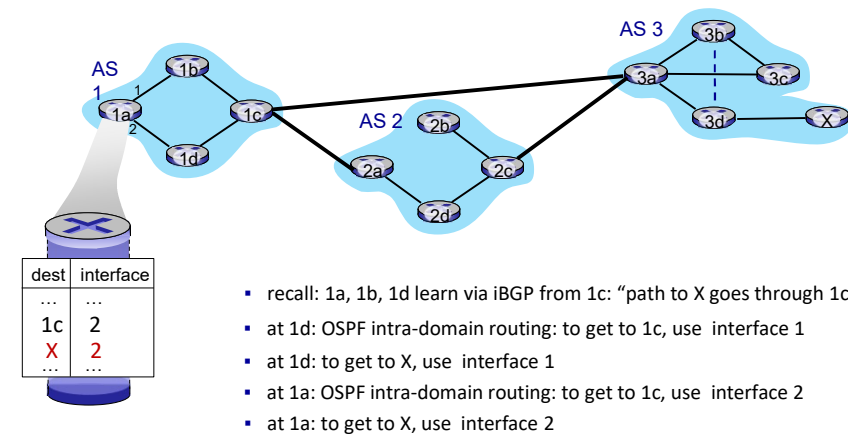
- BGP messages exchanged between peers over TCP connection
- BGP messages:
 - **OPEN**: opens TCP connection to remote BGP peer and authenticates sending BGP peer
 - **UPDATE**: advertises new path (or withdraws old)
 - **KEEPALIVE**: keeps connection alive in absence of UPDATES; also ACKs OPEN request
 - **NOTIFICATION**: reports errors in previous msg; also used to close connection

BGP path advertisement



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BGP path advertisement



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Why different Intra-, Inter-AS routing ?

policy:

- inter-AS: admin wants control over how its traffic routed, who routes through its network
- intra-AS: single admin, so policy less of an issue

scale:

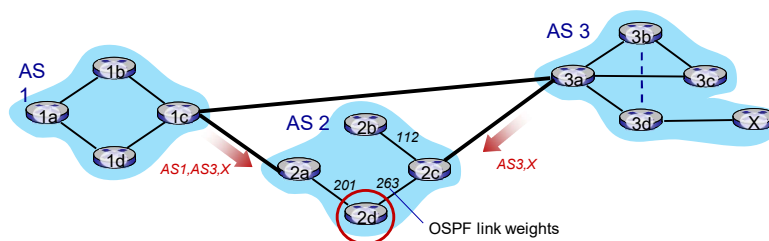
- hierarchical routing saves table size, reduced update traffic

performance:

- intra-AS: can focus on performance
- inter-AS: policy dominates over performance

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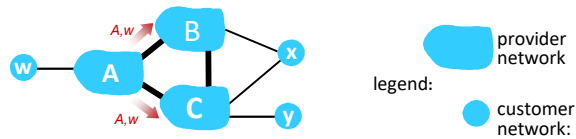
Hot potato routing



- 2d learns (via iBGP) it can route to X via 2a or 2c
- **hot potato routing**: choose local gateway that has least *intra-domain* cost (e.g., 2d chooses 2a, even though more AS hops to X): don't worry about inter-domain cost!

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BGP: achieving policy via advertisements

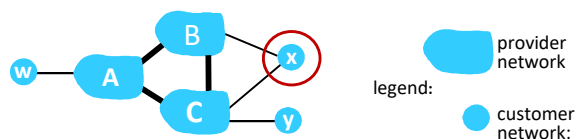


ISP only wants to route traffic to/from its customer networks (does not want to carry transit traffic between other ISPs – a typical “real world” policy)

- A advertises path Aw to B and to C
- B *chooses not to advertise* BAw to C!
 - B gets no “revenue” for routing CBAw, since none of C, A, w are B’s customers
 - C does *not* learn about CBAw path
- C will route CAw (not using B) to get to w

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BGP: achieving policy via advertisements (more)



ISP only wants to route traffic to/from its customer networks (does not want to carry transit traffic between other ISPs – a typical “real world” policy)

- A,B,C are **provider networks**
- x,w,y are **customer** (of provider networks)
- x is **dual-homed**: attached to two networks
- **policy to enforce**: x does not want to route from B to C via x
 - .. so x will not advertise to B a route to C

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BGP route selection

- router may learn about more than one route to destination AS, selects route based on:
 1. local preference value attribute: policy decision
 2. shortest AS-PATH
 3. closest NEXT-HOP router: hot potato routing
 4. additional criteria



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Characteristic of the path

- ORIGIN
 - Source of the information (IGP/EGP/incomplete)
- AS_PATH
- NEXT_HOP
- MED (MULTI_EXIT_DISCRIMINATOR)
- LOCAL_PREF
- ATOMIC_AGGREGATE
- AGGREGATOR
- COMMUNITY



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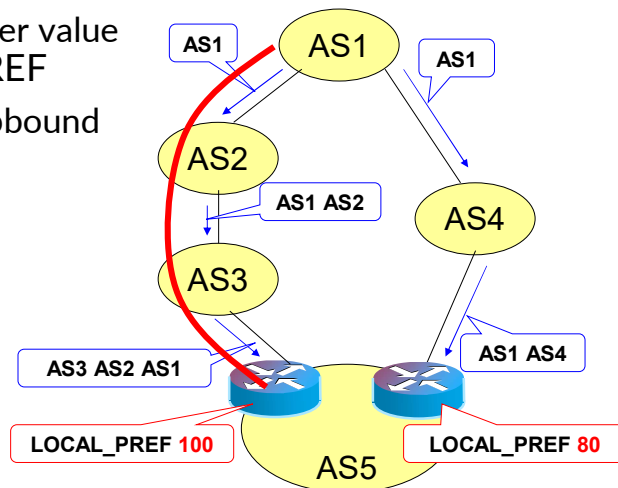
Select the path

- Step 1: Compare LOCAL_PREF
- Step 2: Compare AS_PATH
- Step 3: Compare ORIGIN
- Step 4: Compare MED
- Step 5: Compare EBGP/IBGP
- Step 6: Compare cost to NEXT_HOP
- Step 7: Compare Router ID

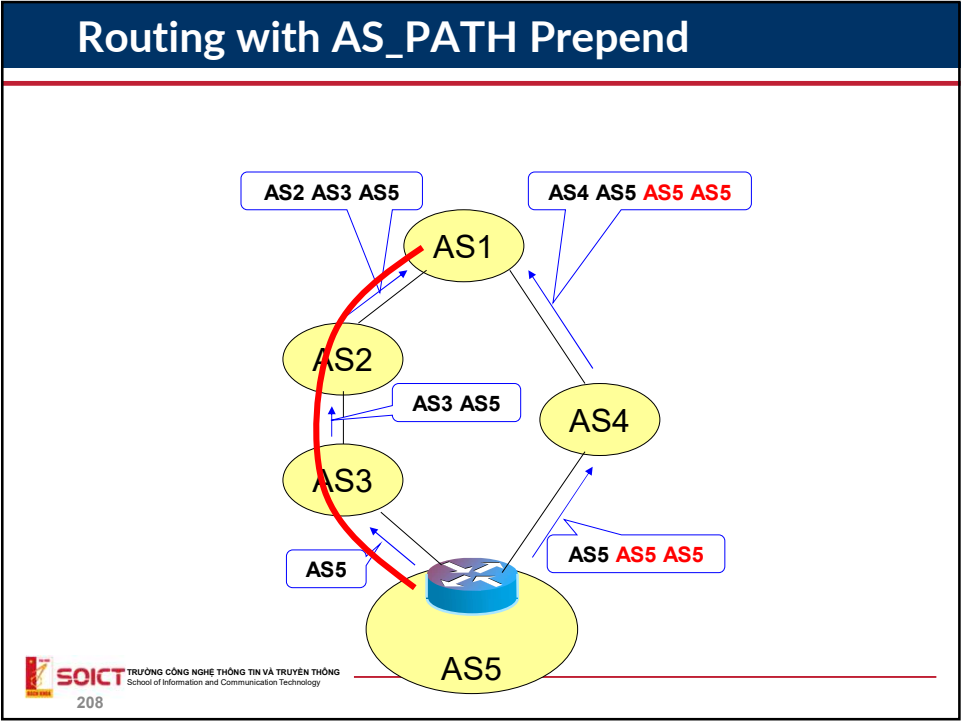
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Sử dụng LOCAL_PREF

- Select the larger value of LOCAL_PREF
- Control the upbound bandwidth



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Example of AS PATH

Network	Next Hop	Metric	LocPrf	Weight	Path
4.79.201.0/26	203.178.136.29	700	500	0	7660 22388 11537 10886 40220
	203.178.136.29	700	500	0	7660 22388 11537 10886 40220
	203.178.136.29	700	500	0	7660 22388 11537 10886 40220
6.1.0.0/16	203.178.136.29	700	500	0	7660 22388 11537 668
	203.178.136.29	700	500	0	7660 22388 11537 668
	203.178.136.29	700	500	0	7660 22388 11537 668
6.2.0.0/22	203.178.136.29	700	500	0	7660 22388 11537 668

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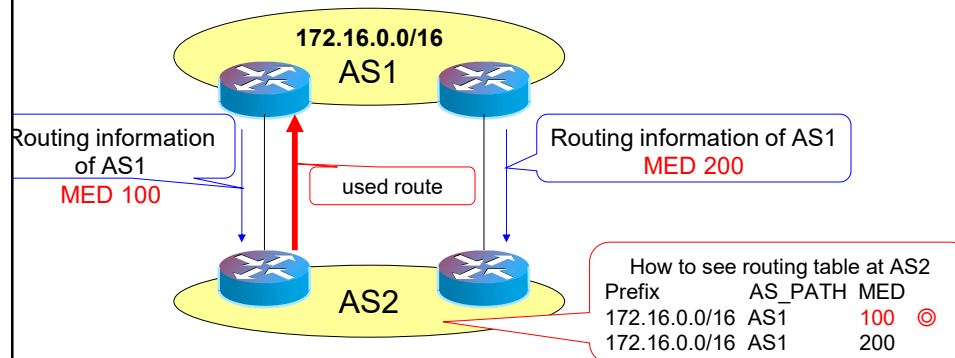
Example of AS PATH prepend

Network	Next Hop	Metric	LocPrf	Weight	Path
8.5.192.0/22	203.178.136.14	100	0	2516	209 13989 13989 13989 13989
	203.178.136.14	100	0	2516	209 13989 13989 13989 13989
	203.178.136.14	100	0	2516	209 13989 13989 13989 13989
8.5.196.0/24	203.178.136.14	100	0	2516	209 13989 13989 13989 13989
	203.178.136.14	100	0	2516	209 13989 13989 13989 13989
	203.178.136.14	100	0	2516	209 13989 13989 13989 13989
8.5.200.0/22	203.178.136.14	100	0	2516	209 13989 13989 13989 13989
	203.178.136.14	100	0	2516	209 13989 13989 13989 13989
	203.178.136.14	100	0	2516	209 13989 13989 13989 13989

Some AS are repeated on the path to make it longer and not being selected

Routing with MED

- In case of 2 AS with many links
- Choose smaller MED
- Apply in controlling bandwidth



Load balancing with MED

- Set MED for different paths
- Also control the bandwidth

